

GROUP A – DATA RUSH
CMPG 321

Group A Members (Data Rush):

GitHub Link: https://github.com/christophvj/DataRush

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Introduce members of the group

Names, surnames and student numbers

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- Luan du Plessis, 46251960
- Pieta Hattingh, 49019767
- Sune Muller, 45926808
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Photos

- Christoph - A fun fact about him: He plays guitar in a band.



- Luan - A fun fact about him: He is very competitive and played 5 sports.



- Pieta – A fun fact about him: He has a somewhat mild obsession with tea.



- Sune - A fun fact about her: She toured 9 different countries in one year.



- Zoe - A fun fact about her: Her favourite colour is pink, favourite animal is a butterfly and preferred hobby is reading.



Phase 1

1.Area of Focus

Rider Purchasing Patterns & Payment Channel Preferences

Objective/Focus

The focus of this project is to analyse rider purchasing patterns and payment channel preferences within Uber and Bolt operations across major South African metropolitan areas.

The analysis will investigate how riders spend money on transport services by examining factors such as trip fares, trip distances, ride categories, tips and other fare-related costs. This will help identify differences in rider spending behaviour across cities, trip types and other relevant categories contained in the dataset.

A further objective is to investigate the payment methods preferred by riders when using Uber and Bolt services. The analysis will compare the adoption and usage of different payment channels and determine whether certain payment methods are associated with higher or lower rates of successful trip completion.

The project will also examine whether relationships exist between payment methods and unsuccessful transactions or trip cancellations. This may help identify payment channels that experience higher failure rates or that are associated with increased cancellations.

Rider tipping behaviour will also be analysed to determine how frequently riders provide tips and whether tip frequency or tip value differs according to trip distance, fare amount, location, ride category or other relevant factors.

The findings of this analysis can support business and operational decision making by helping Uber and Bolt better understand rider behaviour, preferred payment channels and spending patterns. These insights could assist the platforms in improving payment options, reducing payment-related friction, developing targeted promotions and improving the overall rider experience.

The primary datasets relevant to this focus area include sa_riders.csv, trip_headers.csv and trip_fare_breakdown.csv. Other related datasets may also be incorporated where necessary to provide additional context and support the analysis.

2. RESEARCH QUESTIONS

The following research questions were developed to support the selected focus area, which is the Rider Purchasing Patterns & Payment Channel Preferences. The results of these questions will be used for consumer and business insights that could help in making decisions regarding pricing, payment channels, and services. Each question is designed to be answerable using the provided urban transport dataset and can subsequently be investigated through SQL queries in Phase 2.

RQ1 – Riders' Provincial Spending Patterns

What is the average spend per rider per completed trip in each of the provinces of South Africa and how do they differ?

This question will identify differences in the riders' spending behaviors between the different provinces. With this analysis, the average amount spent on completed trips can be compared. This will determine whether the riders spend more or less per trip in the different provinces. From a business perspective, the research could help define valuable areas for the business and understand which locations rider spending is maximum.

RQ2 – Trip Distance Distribution

How do trip distance distributions for completed trips vary across South African provinces and ride categories?

This question will investigate whether riders in various provinces tend to make shorter or longer trips. The results could give insight into how riders use transport services in different places and cover the Budget, Standard, Comfort and Large Capacity ride types. From a business perspective, the results could give insights into the usage of different transport by riders in different provinces.

RQ3 – Distance and Spending

What is the relationship between the trip distance, the trip duration and the total fare paid by the riders across the different ride categories?

This question focuses on identifying a connection between characteristics of a ride, its cost and how much riders pay for it. The analysis will highlight the average distance and duration of trips in different categories and uncover the relationship between these characteristics and the rider spending. From a business perspective, the insights could be useful for both consumers and businesses to understand factors affecting ride cost and rider spending.

RQ4 – Tip Frequency

How does tip frequency and average tip amount differ across South African provinces and ride categories?

This question investigates how much the tip frequency and the average tip amount differs in the different South African provinces. From a business perspective, the result could give insight into where most people tip, how often and how much the person will tip on average.

RQ5 – Payment Method Adoption

What are the most frequently used payment methods among riders, and how does the uptake of different payment methods vary between the provinces of South Africa?

This question investigates the different payments channels - how and where they are used. The comparison of Cash, Credit Card, Debit Card, EFT/Ozow and In-App Wallet payments will reveal if the riders from different provinces have a different payment behavior. From a business perspective, the results may be used for making decisions regarding the choice of payment channels, digital payments adoption, and other factors related to payment options that can be developed further.

RQ6 – Payment Preferences

To what extent is the rider's actual mode of payment different from their preferred mode of payment?

This question investigates whether riders generally use the payment method they identify as their preferred option or does their actual payment behavior differ from their stated preference. It will help to understand the rider's payment behavior in more detail by comparing their preferred payment method with the one they use in reality. From a business perspective, the results will be beneficial for the company when making decisions about which payment channel to develop further.

RQ7 – Payment Method and Trip Cancellations

Is there a relationship between the payment method and the likelihood of a rider-initiated or driver-initiated cancellation of a trip?

This question compares payment methods against trip outcomes to determine whether certain payment channels are associated with different cancellation patterns. Through the analysis, it will be possible to see whether there is any payment method that leads to more frequent trip cancellations. This could provide operational insight into potential patterns surrounding payment choice and unsuccessful trips. This allows the platform to identify the areas that will need further investigations.

RQ8 – Surge Pricing and Rider Spending

How does the surge pricing affect the riders spending across different ride categories and payment methods?

This question will investigate the effects of change in the value of the multiplier of surge prices in relation to the amount of money paid by the riders for their completed trips. The results will provide strategic business insight into the relationship between dynamic pricing and rider expenditure while also providing consumer insight into how surge conditions affect trip costs.

2.1 Alignment of Research Questions with the Focus Area

Question	Focus	Important data
RQ1	Spending	Province, city, total fare
RQ2	Distance distribution	Distance, province, category
RQ3	Spending determinants	Distance, duration, fare
RQ4	Tipping	Tips, province, category
RQ5	Payment adoption	Payment method, province
RQ6	Payment preferences	Preferred payments method, actual payment method.
RQ7	Payment + cancellations	Payment method, trip status
RQ8	Surge + spending	Surge multiplier, fare, category, payment

Together, these questions address the key components of Rider Purchasing Patterns & Payment Channel Preferences. RQ1 to RQ4 focuses on the riders' purchasing behaviors by examining geographical spending, trip distances, trip characteristics and tipping. RQ5 to RQ7 investigates the payment method behavior by considering the payment adoption, preferred versus the actual payment methods and cancellation patterns. Lastly RQ8 studies the connection between the rider's purchasing behavior and the payment channel with the rising price.

The questions are structured so that they can later be translated into SQL queries involving join operations, filtering, grouping and aggregation. More advanced SQL techniques can be incorporated where appropriate during Phase 2.

3 Dataset Description

3.1 Overview of the Uber Bolt dataset

The provided urban transport dataset represents Uber Bolt activity in South Africa, consisting of the following structured CSV files.

- sa_riders.csv
- sa_drivers.csv
- vehicles.csv
- pricing_surge_zones.csv
- trip_headers.csv
- trip_fare_breakdown.csv
- trip_reviews.csv

The CSV files mentioned above contains information about riders, drivers, vehicles, pricing zones, trips, fares, and reviews.

For the project's selected area of focus, "**Rider Purchasing Patterns & Payment Channel Preferences**," the following CSV dataset files are identified as most relevant.

1. sa_riders.csv
2. trip_headers.csv
3. trip_fare_breakdown.csv
4. pricing_surge_zones.csv

trip_reviews.csv may also provide supporting information if required later during the analysis.

3.2 sa_riders.csv

The sa_riders.csv dataset contains information about 1200 riders with the following fields.

Field	Data Type	Description	Relevance
rider_id	String/Text	Unique rider identifier	Primary identifier for riders
first_name	String/Text	Rider's first name	Not required for analysis
last_name	String/Text	Rider's surname	Not required for analysis
mobile_number	String	Rider mobile number	Not required for analysis
home_city	String/Text	Rider's home city	Relevant for geographic analysis
province	String/Text	Rider's province	Highly relevant
preferred_payment	String/Text	Rider's preferred payment method	Highly relevant
account_created_date	Date/Text	Date the rider account was created	Potential supporting variable

3.2.1 Relevant domains

The home_city field contains the following cities:

- Cape Town
- Durban
- Gqeberha
- Johannesburg
- Potchefstroom
- Pretoria
- Vanderbijlpark

The province field contains:

- Eastern Cape
- Gauteng
- KwaZulu-Natal
- North West
- Western Cape

The preferred_payment field contains five payment methods:

- CASH
- CREDIT_CARD
- DEBIT_CARD
- EFT_OZOW
- IN_APP_WALLET

The rider_id field also serves an important role because it allows rider information to be connected to individual trips in trip_headers.csv.

3.3 trip_headers.csv

The trip_headers.csv dataset contains information about 50000 trip records and represents the core operational transaction information. The following fields are identified.

Field	Data Type	Description	Relevance
trip_id	String/Text	Unique trip identifier	Primary trip identifier
rider_id	String/Text	Identifier of the rider	Links trips to riders
driver_id	String/Text	Identifier of the driver	Supporting relationship
zone_id	String/Text	Identifier of the transport zone	Relevant to geographic/pricing analysis
ride_category	String/Text	Type of ride selected	Highly relevant
request_timestamp	Date/Time	Date and time the trip was requested	Relevant to temporal analysis
trip_status	String/Text	Outcome of the trip	Highly relevant
cancellation_reason	String/Text	Reason for cancellation	Relevant to cancellation analysis

3.3.1 Relevant Domains

The ride_category field contains the following categories:

- Budget (Uber Go/Bolt Go)
- Standard (UberX/Bolt)
- Comfort (Uber Comfort)
- Large Capacity (UberXL/Bolt XL)

The trip_status field consists of the following statuses:

- COMPLETED
- CANCELLED_BY_RIDER
- CANCELLED_BY_DRIVER

The cancellation_reason field contains the following cancellation reasons:

- CHANGED_MIND
- DRIVER_TOO_FAR
- PRICING_SURGE
- RIDER_NO_SHOW
- SAFETY_CONCERN
- TRAFFIC_DELAY

3.4 trip_fare_breakdown.csv

The trip_fare_breakdown.csv dataset contains the financial information associated with individual trips in the supplied South African Uber/Bolt Urban Transport dataset. It is one of the most relevant datasets for the selected focus area of Rider Purchasing Patterns and Payment Channel Preferences, because it contains information relating directly to rider expenses, trip distance, tipping and the payment channel used for each trip.

Each row represents a single record, and each column represents a specific data field. The first row contains the field names. The file contains 50,000 records and 11 fields, with each record representing the fare breakdown associated with one trip. The data contained in the file is a combination of identifier, numerical, monetary, and categorical data.

Field	Data Type	Range / Domain	Description	Relevance to Area of Focus
trip_id	String/Text	50,000 unique trip identifiers	Unique identifier for each trip	Links fare information to the corresponding trip (trip_header)
distance_km	Decimal	0.00–48.00 km	Distance associated with the trip in kilometres	Highly relevant to trip-distance analysis
duration_minutes	Integer	0–182 minutes	Duration associated with the trip	Supporting trip characteristic
surge_multiplier	Decimal	1.0–2.4	Pricing multiplier applied to the trip	Relevant to differences in rider expenditure
base_fare_zar	Decimal	R20.00–R35.00	Starting fare associated with the trip	Relevant to trip pricing
tolls_zar	Decimal	R0.00–R42.00	Toll charges associated with the trip	Supporting component of rider expenditure

tip_zar	Decimal	R0.00–R50.00	Tip amount associated with the trip	Highly relevant to tip-frequency analysis
total_fare_zar	Decimal	R0.00–R1,988.22	Total fare associated with the trip	Highly relevant to rider spending analysis
platform_commission_zar	Decimal	R0.00–R494.59	Amount retained by the platform	Supporting financial variable
driver_payout_zar	Decimal	R0.00–R1,550.11	Amount allocated to the driver	Supporting financial variable
payment_method	String/Text	CASH, CREDIT_CARD, DEBIT_CARD, EFT_OZOW, IN_APP_WALLET	Payment channel used for the trip	Highly relevant to payment-channel preference analysis

3.4.1 Numerical domains

The trip_id field acts as the main identifier in the dataset. All 50,000 trip_id values are unique. Each fare record can be distinguished from every other record. The uniqueness of the field also makes it a suitable candidate to be used as a primary key when it is used in a relational database table.

The dataset contains no missing values across the eleven fields. But it does contain zero values within some numerical and monetary fields. A zero value should therefore not automatically be interpreted as missing data.

The distance_km field contains decimal values ranging from 0.00 km to 48.00 km. This field aligns with the focus area since the project requires the analysis of trip-distance distributions. It provides a measurable representation of the distance associated with each trip and can later be compared with spending, tipping, and payment behaviour.

The duration_minutes field ranges from 0 to 182 minutes. It provides useful supporting information about the characteristics of each trip and may help to place differences in trip fares into context.

The dataset contains several monetary fields. base_fare_zar represents the starting fare and ranges from R20.00 to R35.00, while tolls_zar records additional toll-related costs and contains values between R0.00 and R42.00. These fields provide additional information about the different cost components contributing to a trip.

The total_fare_zar field ranges from R0.00 to R1,988.22 and is an important field for the selected area of focus. It represents the overall fare associated with an individual trip and provides the main monetary variable for examining rider spending habits.

The tip_zar field contains the values R0.00, R10.00, R15.00, R20.00 and R50.00. The presence of both zero and positive values makes the field suitable for investigating tip frequency as well as tip amounts. In later analysis, this field can be used to determine how often riders tip and whether tipping patterns vary between different categories of riders or trips.

The surge_multiplier field contains values ranging from 1.0 to 2.4. It provides additional pricing context because different multiplier values may contribute to differences in the total fare associated with otherwise similar trips.

The payment_method field is particularly important, its domain consists of five payment channels, CASH, CREDIT_CARD, DEBIT_CARD, EFT_OZOW, IN_APP_WALLET.

This means that the dataset contains both traditional cash transactions and several electronic payment mechanisms. The field identifies the payment method associated with each trip and can later be compared with fare values, trip distances, tips, and trip outcomes.

The final two monetary fields, platform_commission_zar and driver_payout_zar, range from R0.00 to R494.59 and R0.00 to R1,550.11. These fields are less directly related to rider purchasing behaviour, but they provide additional detail regarding how the financial value associated with a trip is distributed.

Overall, trip_fare_breakdown.csv can be classified as a transaction-level financial dataset. The fields most relevant to the selected area of focus are distance_km, total_fare_zar, tip_zar and payment_method, while fields such as surge_multiplier, base_fare_zar, tolls_zar and duration_minutes provide supporting information that can help explain differences in rider expenditure and behaviour.

3.5 pricing_surge_zones.csv

The pricing_surge_zones dataset contains the geographical pricing information for the different urban transport zones represented in the supplied South African Uber/Bolt dataset. It provides information about where the pricing zones are located and the tariff associated with each zone. It therefore provides useful geographical and pricing context for the group's selected area of focus.

Each row represents one pricing zone and each column represents a specific characteristic of that zone. The file contains 9 records and 6 fields.

Unlike trip_fare_breakdown, which contains a separate record for individual trip transactions, pricing_surge_zones contains information that describes stable geographical pricing rules. The dataset can therefore be classified as reference or master data. It contains a combination of identifier data, categorical geographical data, and monetary tariff data.

Field	Data Type	Range / Domain	Description	Relevance to Area of Focus
zone_id	String/Text	9 unique identifiers	Unique identifier for each pricing zone	Links pricing-zone data to trip information
zone_name	String/Text	9 zone names	Descriptive name of the geographical pricing zone	Provides geographical context
city	String/Text	7 South African cities	City in which the pricing zone is located	Relevant to geographical comparison
base_fare_zar	Decimal	R20.00–R35.00	Standard starting fare for the zone	Relevant to rider expenditure
per_km_rate_zar	Decimal	R8.00–R12.00 per km	Charge applied per kilometre travelled	Highly relevant to distance and pricing
per_min_rate_zar	Decimal	R1.00–R1.60 per minute	Charge applied per minute of travel	Relevant to pricing context

3.5.1 Numerical domains

The zone_id field uniquely identifies each of the nine pricing-zone records. All nine zone_id values are unique, and the field follows a text-based identifier format such as ZONE_JHB_CBD, ZONE_SANDTON, and ZONE_POTCH_NWU. Because each value uniquely represents one pricing zone, zone_id is suitable for use as the primary key when the dataset is later implemented as a database table.

The zone_id field also occurs in trip_headers. This enables individual trips to be related to the geographical pricing zone associated with them. The relationship is important because the pricing data stored in this file does not describe a single individual trip. Instead, it provides the pricing context that can be associated with many trip records.

There are no missing values across the nine records and six fields in the dataset.

The city field has a domain consisting of seven South African cities; Johannesburg, Pretoria, Vanderbijlpark, Cape Town, Durban, Potchefstroom, Gqeberha, these cities correspond directly with the geographical areas specified for the Urban Transport dataset.

The nine individual pricing zones represented in the dataset are:

Zone Name	City	Base Fare	Per-km Rate	Per-minute Rate
Johannesburg CBD / Maboneng	Johannesburg	R25.00	R9.50	R1.20
Sandton Gautrain & Financial Core	Johannesburg	R28.00	R11.00	R1.50
Menlyn Maine & Hatfield	Pretoria	R24.00	R9.00	R1.10
Vanderbijlpark Vaal Mall & VUT Campus	Vanderbijlpark	R21.00	R8.50	R1.05
V&A Waterfront & Atlantic Seaboard	Cape Town	R30.00	R12.00	R1.60
Cape Town International Airport	Cape Town	R35.00	R11.50	R1.40
Umhlanga Rocks & Gateway	Durban	R26.00	R10.00	R1.30
NWU Campus & Bult Area	Potchefstroom	R20.00	R8.00	R1.00
Summerstrand & Airport	Gqeberha	R22.00	R8.50	R1.05

Johannesburg and Cape Town each contain two separate pricing zones, while Pretoria, Vanderbijlpark, Durban, Potchefstroom, and Gqeberha each contain one.

The base_fare_zar field is associated with each pricing zone. Values range from R20.00 to R35.00. The lowest base fare is recorded for the NWU Campus & Bult Area in Potchefstroom, while the highest is recorded for Cape Town International Airport. The range demonstrates that the starting fare is not identical across all geographical locations.

The per_km_rate_zar field contains the distance-based component of the pricing structure. Values range from R8.00 to R12.00 per kilometre. The amount associated with travelling a given distance may differ depending on the geographical zone in which the trip takes place.

Although the file is named pricing_surge_zones, the dataset itself does not contain a field storing the surge multiplier applied to an individual trip. Instead, it contains the standard geographical pricing components of base fare, rate per kilometre and rate per minute. The actual trip-specific surge_multiplier is stored in trip_fare_breakdown. The geographical tariff structure and individual trip surge multiplier should therefore be treated as separate pieces of pricing information.

Overall, pricing_surge_zones.csv can be classified as a geographical pricing reference dataset. The most relevant fields for the group's focus are city, zone_name, base_fare_zar, per_km_rate_zar and per_min_rate_zar, because these fields provide the location and tariff information required to place rider spending and trip-distance patterns into geographical context.

3.6 trip_reviews.csv

The trip_reviews.csv dataset contains 25,055 records and holds the satisfaction layer of the urban transport data. Each record captures the rating exchanged between a rider and a driver after a trip, together with a comment.

Field	Data type	Format / Domain	Description and relevance
review_id	Text	Pattern REV_0000001 - REV_0049999	Primary key. Sequence is non-contiguous, as numbering follows the underlying trip.
trip_id	Text	Matches trip_headers. trip_id; unique within this file	Foreign key. The only join path to trip, fare and rider data.
rider_rating_of_driver	Integer	Ordinal, 1 - 5 (all five values present; mean 3.65)	Highly relevant. Permits satisfaction to be tested against tipping and payment channel.
driver_rating_of_rider	Integer	Ordinal, 3 - 5 only (mean 4.51; no 1 or 2 ratings occur)	Supporting. Rider side behaviour indicator, but with some limited variation.
feedback_comment	Text	Only 6 distinct values, 29 - 44 characters, no blanks	Supporting. Basically, a coded category, rather than free text.

3.6.1 Key Observations

- No missing values in any field.
- Both rating fields use ordinal integers, not continuous numbers. Averages have to be read against a fixed scale.
- feedback_comment is a data-quality issue. It is stored as text but draws from only six set sentences, each used about 4,100 times. So, it acts like a categorical variable, has no payment-related commentary, and is not useful for genuine text analysis.
- No rider or driver identifiers exist. trip_id is the only way to link these reviews to the rest of the dataset.
- Reviews only exist for completed trips. All 25,055 reviews match a completed trip. None of the 16,669 cancelled trips have a review.
- Of the 33,331 completed trips, 75.2% have a review, leaving 8,276 completed trips unreviewed.
- Any join from trips to reviews must be a LEFT OUTER JOIN. An inner join would drop cancelled trips and a quarter of completed ones, which would distort spending calculations.
- Every trip_id in this file does exist in the trip_headers table.
- This file has no monetary or payment fields, so it is a supporting dataset, not a primary one.
- It adds a satisfaction score that no other file provides. This lets the team test if rider ratings vary with tip behaviour, payment channel, or surge level.
- Early looks show only small differences. Average rider ratings fall between 3.62 and 3.66 across all five payment methods

3.7 Dataset Relationships

- The seven files connect through two identifiers: rider_id and trip_id.
- trip_headers.csv sits at the centre of the structure.
- It is the only file that holds rider, driver, and zone identifiers together.
- So, every analysis path has to go through this file.

File	Primary key	Foreign key	Relationship and relevance
sa_riders.csv	rider_id	None	1:M to trip_headers
trip_headers.csv	trip_id	rider_id, driver_id, zone_id	M:1 to sa_riders M:1 to sa_drivers M:1 to pricing_surge_zones
trip_fare_breakdown.csv	trip_id	trip_id	1:1 with trip_headers
pricing_surge_zones.csv	zone_id	none	1:M to trip_headers
trip_reviews.csv	review_id	trip_id	M:1 to trip_headers
sa_drivers.csv	driver_id	None	1:M to trip_headers
vehicles.csv	vehicle_id	driver_id	1:1 with sa_drivers

Phase 2

4. Database Design and Setup

The seven CSV files that came with the dataset were represented as seven relational tables, each one-to-one with the seven source files, so that the data can be bulk loaded with minimal transformation. Table and Column names are original CSV names/headers. The design below is written in MySQL SQL syntax and was implemented using MySQL as the relational database management system (RDBMS).

4.1 Design Principles

- Tables are created in parent before child order and dropped in the reverse order since MySQL will not create or drop a table out of order relative to its foreign keys.
- Every table has a single-column primary key that uniquely identifies each record.
- Foreign keys enforce every relationship identified in Section 3.7.
- Unique constraints are added where the data confirms a 1:1 relationship or a naturally unique business key such as a mobile number and vehicle license plates.
- NOT NULL is applied to every column that is fully populated in the provided data, so incomplete records cannot be inserted.
- CHECK constraints are applied to all categorical domains on every column with a fixed set of valid values such as preferred_payment, platform_affiliation, ride_category, trip_status and payment_method. This prevents invalid or misspelled category values from being stored.
- CHECK constraints enforce sensible numeric ranges such as restricting ratings to 1-5, and vehicle year is restricted to 2000-2026 catching data entry errors that a plain numeric type wouldn't.
- CHECK constraints also enforce non-negative values on every quantity column.

4.2 Table Creation & Constraints – MySQL

The tables are created in the order below, so that referenced parent tables always exist before the child tables that carry a foreign key to them.

4.2.1 sa_riders

```
32      -- ============================================================
33      -- Create sa_riders Table
34      -- ============================================================
35      CREATE TABLE sa_riders (
36          rider_id                VARCHAR(12) PRIMARY KEY,
37          first_name              VARCHAR(50) NOT NULL,
38          last_name               VARCHAR(50) NOT NULL,
39          mobile_number           VARCHAR(15) NOT NULL UNIQUE,
40          home_city               VARCHAR(30) NOT NULL,
41          province                VARCHAR(30) NOT NULL,
42          preferred_payment       VARCHAR(20) NOT NULL,
43          account_created_date    DATE        NOT NULL,
44
45          CONSTRAINT chk_riders_preferred_payment CHECK (
46              preferred_payment IN ('CASH','CREDIT_CARD','DEBIT_CARD','EFT_OZOW','IN_APP_WALLET')
47          )
48      );
```

Figure 1sa_riders table SQL

4.2.2 sa_drivers

```sql
-- ============================================================
-- Create sa_drivers Table
-- ============================================================
CREATE TABLE sa_drivers (
    driver_id                   VARCHAR(10)     PRIMARY KEY,
    first_name                  VARCHAR(50)     NOT NULL,
    last_name                   VARCHAR(50)     NOT NULL,
    operating_city              VARCHAR(30)     NOT NULL,
    platform_affiliation        VARCHAR(25)     NOT NULL,
    platform_commission_pct     DECIMAL(4,2)    NOT NULL,
    driver_rating               DECIMAL(3,2)    NOT NULL,
    total_lifetime_trips        INT             NOT NULL,
    onboarding_date             DATE            NOT NULL,

    CONSTRAINT chk_drivers_platform CHECK (
        platform_affiliation IN ('Uber','Bolt','Dual-Platform (Both)')
    ),
    CONSTRAINT chk_drivers_commission CHECK (
        platform_commission_pct BETWEEN 0 AND 1
    ),
    CONSTRAINT chk_drivers_rating CHECK (
        driver_rating BETWEEN 1 AND 5
    ),
    CONSTRAINT chk_drivers_lifetime_trips CHECK (
        total_lifetime_trips >= 0
    )
);
```

Figure 2 sa_drivers table SQL

4.2.3 vehicles

```
80      -- ============================================================
81      -- Create vehicles Table
82      -- ============================================================
83      CREATE TABLE vehicles (
84          vehicle_id      VARCHAR(10) PRIMARY KEY,
85          driver_id       VARCHAR(10) NOT NULL UNIQUE,
86          make            VARCHAR(30) NOT NULL,
87          model           VARCHAR(30) NOT NULL,
88          year            SMALLINT    NOT NULL,
89          license_plate   VARCHAR(15) NOT NULL UNIQUE,
90          category_type   VARCHAR(40) NOT NULL,
91
92          CONSTRAINT fk_vehicles_driver FOREIGN KEY (driver_id)
93              REFERENCES sa_drivers(driver_id),
94          CONSTRAINT chk_vehicles_year CHECK (
95              year BETWEEN 2000 AND 2026
96          )
97      );
98
```

Figure 3 vehicles table SQL

4.2.4 pricing_surge_zones

```
100     -- ============================================================
101     -- Create pricing_surge_zones Table
102     -- ============================================================
103     CREATE TABLE pricing_surge_zones (
104         zone_id             VARCHAR(25)     PRIMARY KEY,
105         zone_name           VARCHAR(60)     NOT NULL,
106         city                VARCHAR(30)     NOT NULL,
107         base_fare_zar       DECIMAL(6,2)    NOT NULL,
108         per_km_rate_zar     DECIMAL(6,2)    NOT NULL,
109         per_min_rate_zar    DECIMAL(6,2)    NOT NULL,
110
111         CONSTRAINT chk_zones_base_fare CHECK (base_fare_zar >= 0),
112         CONSTRAINT chk_zones_per_km CHECK (per_km_rate_zar >= 0),
113         CONSTRAINT chk_zones_per_min CHECK (per_min_rate_zar >= 0)
114     );
115
```

Figure 4 pricing_surge_zones table SQL

4.2.5 trip_headers

```sql
-- ============================================================
-- Create trip_headers Table
-- ============================================================
CREATE TABLE trip_headers (
    trip_id                 VARCHAR(20) PRIMARY KEY,
    rider_id                VARCHAR(12) NOT NULL,
    driver_id               VARCHAR(10) NOT NULL,
    zone_id                 VARCHAR(25) NOT NULL,
    ride_category           VARCHAR(40) NOT NULL,
    request_timestamp       DATETIME    NOT NULL,
    trip_status             VARCHAR(25) NOT NULL,
    cancellation_reason     VARCHAR(25),

    CONSTRAINT fk_trips_rider FOREIGN KEY (rider_id)
        REFERENCES sa_riders(rider_id),
    CONSTRAINT fk_trips_driver FOREIGN KEY (driver_id)
        REFERENCES sa_drivers(driver_id),
    CONSTRAINT fk_trips_zone FOREIGN KEY (zone_id)
        REFERENCES pricing_surge_zones(zone_id),
    CONSTRAINT chk_trips_ride_category CHECK (
        ride_category IN (
            'Budget (Uber Go/Bolt Go)','Standard (UberX/Bolt)',
            'Comfort (Uber Comfort)','Large Capacity (UberXL/Bolt XL)'
        )
    ),
    CONSTRAINT chk_trips_status CHECK (
        trip_status IN ('COMPLETED','CANCELLED_BY_RIDER','CANCELLED_BY_DRIVER')
    ),
    CONSTRAINT chk_trips_cancel_reason CHECK (
        cancellation_reason IN (
            'CHANGED_MIND','DRIVER_TOO_FAR','PRICING_SURGE',
            'RIDER_NO_SHOW','SAFETY_CONCERN','TRAFFIC_DELAY'
        ) OR cancellation_reason IS NULL
    )
);
```

Figure 5 trip_headers table SQL

4.2.6 trip_fare_breakdown

```sql
-- ============================================================
-- Create trip_fare_breakdown Table
-- ============================================================
CREATE TABLE trip_fare_breakdown (
    trip_id                 VARCHAR(20)     PRIMARY KEY,
    distance_km             DECIMAL(5,2)    NOT NULL,
    duration_minutes        INT             NOT NULL,
    surge_multiplier        DECIMAL(3,2)    NOT NULL,
    base_fare_zar           DECIMAL(6,2)    NOT NULL,
    tolls_zar               DECIMAL(6,2)    NOT NULL DEFAULT 0,
    tip_zar                 DECIMAL(6,2)    NOT NULL DEFAULT 0,
    total_fare_zar          DECIMAL(9,2)    NOT NULL,
    platform_commission_zar DECIMAL(8,2)    NOT NULL,
    driver_payout_zar       DECIMAL(9,2)    NOT NULL,
    payment_method          VARCHAR(20)     NOT NULL,

    CONSTRAINT fk_fares_trip FOREIGN KEY (trip_id)
        REFERENCES trip_headers(trip_id),
    CONSTRAINT chk_fares_distance CHECK (distance_km >= 0),
    CONSTRAINT chk_fares_duration CHECK (duration_minutes >= 0),
    CONSTRAINT chk_fares_surge CHECK (surge_multiplier >= 1),
    CONSTRAINT chk_fares_base_fare CHECK (base_fare_zar >= 0),
    CONSTRAINT chk_fares_tolls CHECK (tolls_zar >= 0),
    CONSTRAINT chk_fares_tip CHECK (tip_zar >= 0),
    CONSTRAINT chk_fares_total CHECK (total_fare_zar >= 0),
    CONSTRAINT chk_fares_commission CHECK (platform_commission_zar >= 0),
    CONSTRAINT chk_fares_payout CHECK (driver_payout_zar >= 0),
    CONSTRAINT chk_fares_payment_method CHECK (
        payment_method IN ('CASH','CREDIT_CARD','DEBIT_CARD','EFT_OZOW','IN_APP_WALLET')
    )
);
```

Figure 6 trip_fare_breakdown table SQL

4.2.7 trip_reviews

```sql
-- ============================================================
-- Create trip_reviews Table
-- ============================================================
CREATE TABLE trip_reviews (
    review_id                   VARCHAR(15) PRIMARY KEY,
    trip_id                     VARCHAR(20) NOT NULL UNIQUE,
    rider_rating_of_driver      SMALLINT    NOT NULL,
    driver_rating_of_rider      SMALLINT    NOT NULL,
    feedback_comment            VARCHAR(150),

    CONSTRAINT fk_reviews_trip FOREIGN KEY (trip_id)
        REFERENCES trip_headers(trip_id),
    CONSTRAINT chk_reviews_rider_rating CHECK (
        rider_rating_of_driver BETWEEN 1 AND 5
    ),
    CONSTRAINT chk_reviews_driver_rating CHECK (
        driver_rating_of_rider BETWEEN 1 AND 5
    )
);
```

Figure 7 trip_reviews table SQL

4.3 Loading the Data

Once the schema above has been created in the MySQL database management system, the seven CSV files can be bulk loaded directly, because every column name in the data definition language matches the corresponding CSV header. Tables must be loaded in the same parent before child order used to create them, so that foreign-key constraints are never violated during the load:

4.3.1 sa_riders.csv

```
LOAD DATA LOCAL INFILE 'C:/Users/chris/OneDrive/Desktop/uber_bolt_sa_dataset/sa_riders.csv'
INTO TABLE sa_riders
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS
(rider_id, first_name, last_name, mobile_number, home_city, province,
 preferred_payment, @created)
SET account_created_date = STR_TO_DATE(@created, '%Y/%m/%d');
```

Figure 8 import sa_riders.csv

4.3.2 sa_drivers.csv

```
LOAD DATA LOCAL INFILE 'C:/Users/chris/OneDrive/Desktop/uber_bolt_sa_dataset/sa_drivers.csv'
INTO TABLE sa_drivers
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS
(driver_id, first_name, last_name, operating_city, platform_affiliation,
 platform_commission_pct, driver_rating, total_lifetime_trips, @onboard)
SET onboarding_date = STR_TO_DATE(@onboard, '%Y/%m/%d');

```

Figure 9 import sa_drivers.csv

4.3.3 pricing_surge_zones.csv

```
LOAD DATA LOCAL INFILE 'C:/Users/chris/OneDrive/Desktop/uber_bolt_sa_dataset/pricing_surge_zones.csv'
INTO TABLE pricing_surge_zones
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS
(zone_id, zone_name, city, base_fare_zar, per_km_rate_zar, per_min_rate_zar);

```

Figure 10 import pricing_surge_zones.csv

4.3.4 vehicles.csv - (depends on sa_drivers)

```
LOAD DATA LOCAL INFILE 'C:/Users/chris/OneDrive/Desktop/uber_bolt_sa_dataset/vehicles.csv'
INTO TABLE vehicles
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS
(vehicle_id, driver_id, make, model, year, license_plate, category_type);
```

Figure 11 import vehicles.csv

4.3.5 trip_headers.csv - (depends on sa_riders, sa_drivers, pricing_surge_zones)

```
LOAD DATA LOCAL INFILE 'C:/Users/chris/OneDrive/Desktop/uber_bolt_sa_dataset/trip_headers.csv'
INTO TABLE trip_headers
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS
(trip_id, rider_id, driver_id, zone_id, ride_category,
 request_timestamp, trip_status, @cancel)
SET cancellation_reason = NULLIF(@cancel, 'N/A');
```

Figure 12 import trip_headers.csv

4.3.6 trip_fare_breakdown.csv - (depends on trip_headers)

```
LOAD DATA LOCAL INFILE 'C:/Users/chris/OneDrive/Desktop/uber_bolt_sa_dataset/trip_fare_breakdown.csv'
INTO TABLE trip_fare_breakdown
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS
(trip_id, distance_km, duration_minutes, surge_multiplier, base_fare_zar,
 tolls_zar, tip_zar, total_fare_zar, platform_commission_zar,
 driver_payout_zar, payment_method);
```

Figure 13 import trip_fare_breakdown.csv

4.3.7 trip_reviews.csv - (depends on trip_headers)

```
LOAD DATA LOCAL INFILE 'C:/Users/chris/OneDrive/Desktop/uber_bolt_sa_dataset/trip_reviews.csv'
INTO TABLE trip_reviews
FIELDS TERMINATED BY ',' OPTIONALLY ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 ROWS
(review_id, trip_id, rider_rating_of_driver, driver_rating_of_rider, feedback_comment);

```

Figure 14 import trip_reviews.csv

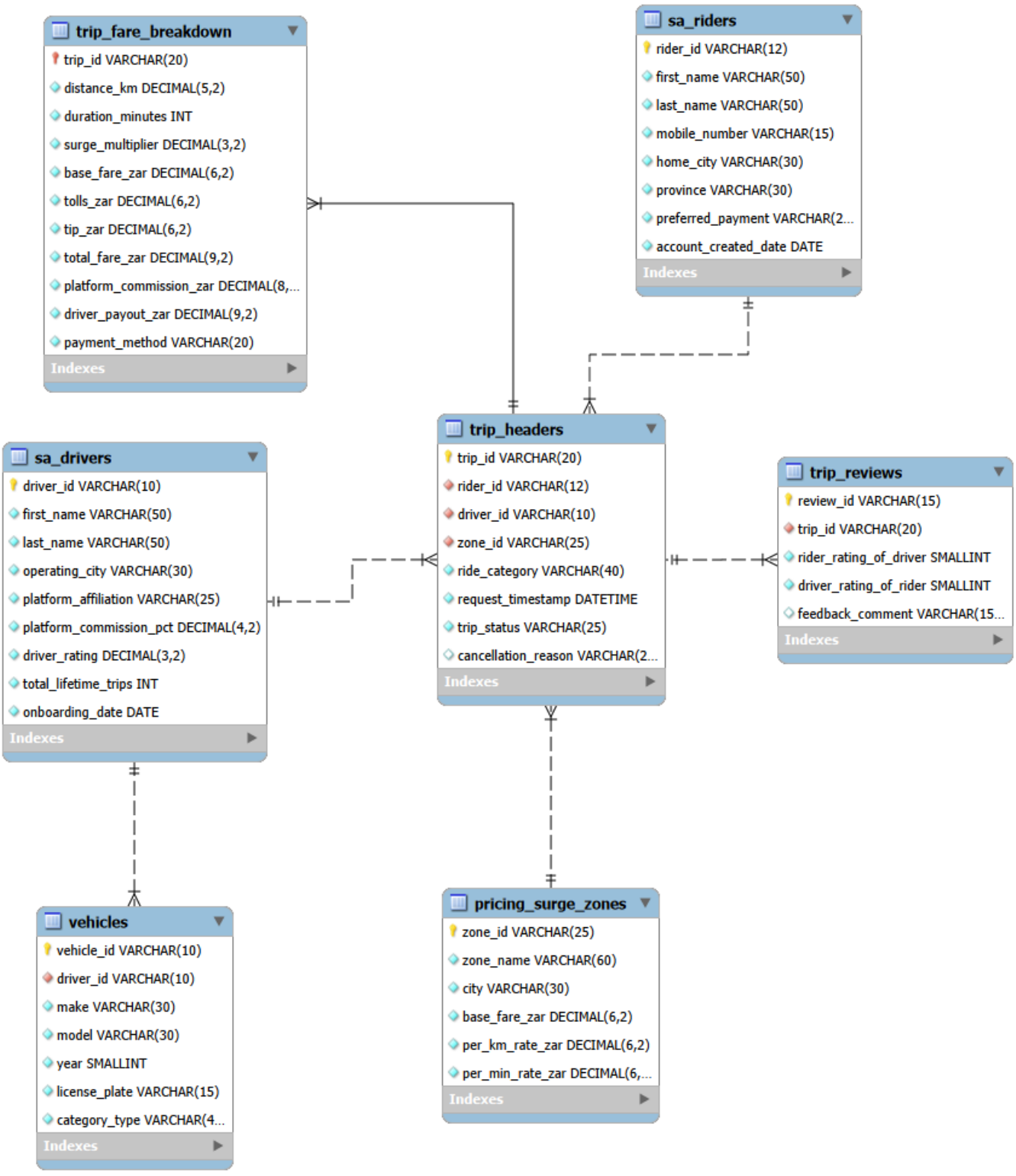
4.3.8 Expected row counts after successful and complete data load:

```
93      SELECT 'sa_riders' t, COUNT(*) row_count FROM sa_riders
94      UNION ALL SELECT 'sa_drivers', COUNT(*) FROM sa_drivers
95      UNION ALL SELECT 'pricing_surge_zones', COUNT(*) FROM pricing_surge_zones
96      UNION ALL SELECT 'vehicles', COUNT(*) FROM vehicles
97      UNION ALL SELECT 'trip_headers', COUNT(*) FROM trip_headers
98      UNION ALL SELECT 'trip_fare_breakdown', COUNT(*) FROM trip_fare_breakdown
99      UNION ALL SELECT 'trip_reviews', COUNT(*) FROM trip_reviews;
```

Figure 15 confirm row counts SQL

Table	Expected Rows
sa_riders.csv	1200
sa_drivers.csv	800
pricing_surge_zones.csv	9
vehicles.csv	800
trip_headers.csv	50000
trip_fare_breakdown.csv	50000
trip_reviews.csv	25055

5. ERD Schema


trip_fare_breakdown
trip_id VARCHAR(20)
distance_km DECIMAL(5,2)
duration_minutes INT
surge_multiplier DECIMAL(3,2)
base_fare_zar DECIMAL(6,2)
tolls_zar DECIMAL(6,2)
tip_zar DECIMAL(6,2)
total_fare_zar DECIMAL(9,2)
platform_commission_zar DECIMAL(8,...
driver_payout_zar DECIMAL(9,2)
payment_method VARCHAR(20)
Indexes
sa_riders
rider_id VARCHAR(12)
first_name VARCHAR(50)
last_name VARCHAR(50)
mobile_number VARCHAR(15)
home_city VARCHAR(30)
province VARCHAR(30)
preferred_payment VARCHAR(2...
account_created_date DATE
Indexes
trip_headers
trip_id VARCHAR(20)
rider_id VARCHAR(12)
driver_id VARCHAR(10)
zone_id VARCHAR(25)
ride_category VARCHAR(40)
request_timestamp DATETIME
trip_status VARCHAR(25)
cancellation_reason VARCHAR(2...
Indexes
sa_drivers
driver_id VARCHAR(10)
first_name VARCHAR(50)
last_name VARCHAR(50)
operating_city VARCHAR(30)
platform_affiliation VARCHAR(25)
platform_commission_pct DECIMAL(4,2)
driver_rating DECIMAL(3,2)
total_lifetime_trips INT
onboarding_date DATE
Indexes
trip_reviews
review_id VARCHAR(15)
trip_id VARCHAR(20)
rider_rating_of_driver SMALLINT
driver_rating_of_rider SMALLINT
feedback_comment VARCHAR(15...
Indexes
vehicles
vehicle_id VARCHAR(10)
driver_id VARCHAR(10)
make VARCHAR(30)
model VARCHAR(30)
year SMALLINT
license_plate VARCHAR(15)
category_type VARCHAR(4...
Indexes
pricing_surge_zones
zone_id VARCHAR(25)
zone_name VARCHAR(60)
city VARCHAR(30)
base_fare_zar DECIMAL(6,2)
per_km_rate_zar DECIMAL(6,2)
per_min_rate_zar DECIMAL(6,...
Indexes

6 SQL Query Development

The following section presents the SQL queries developed to answer the research questions that were identified in Phase 1. The queries use joins, aggregations, subqueries and other SQL techniques to analyse rider spending patterns, trip characteristics, tipping behaviour, payment preferences, cancellations and surge pricing.

6.1 RQ1 Average Spend per Completed Trip by Province

```sql
/*================================================================================
6.1  RQ1 | Average spend per completed trip, by province
--------------------------------------------------------------------------------
Question : What is the average spend per completed trip in each province?
================================================================================ */
SELECT
    r.province,
    COUNT(*)                                        AS completed_trips,
    ROUND(AVG(f.total_fare_zar), 2)                 AS avg_fare_per_trip_zar,
    ROUND(AVG(f.total_fare_zar)
          - SUM(SUM(f.total_fare_zar)) OVER ()
            / SUM(COUNT(*)) OVER (), 2)             AS diff_vs_national_avg_zar,
    ROUND(SUM(f.total_fare_zar), 2)                 AS total_spend_zar,
    RANK() OVER (ORDER BY AVG(f.total_fare_zar) DESC) AS spend_rank
FROM trip_headers t
    INNER JOIN sa_riders r           ON r.rider_id = t.rider_id
    INNER JOIN trip_fare_breakdown f ON f.trip_id  = t.trip_id
WHERE t.trip_status = 'COMPLETED'
GROUP BY r.province
HAVING COUNT(*) >= 30
ORDER BY spend_rank;
```

Figure 16 RQ1 SQL

province	completed_trips	avg_fare_per_trip_zar	diff_vs_national_avg_zar	total_spend_zar	spend_rank
KwaZulu-Natal	4858	465.94	3.03	2263553.17	1
Gauteng	14850	463.47	0.56	6882568.87	2
Eastern Cape	4764	462.94	0.03	2205449.84	3
Western Cape	4617	461.31	-1.60	2129876.83	4
North West	4242	459.19	-3.72	1947882.23	5

Figure 17 RQ1 SQL Result

6.2 RQ2 Trip Distance Distribution by Province & Category

```sql
/*============================================================================
6.2  RQ2 | Trip distances by province and ride category
----------------------------------------------------------------------------
Question: How do trip distances differ between provinces and ride categories?
============================================================================ */
SELECT
    r.province,
    t.ride_category,
    COUNT(*)                          AS completed_trips,
    ROUND(AVG(f.distance_km), 2)      AS avg_distance_km,
    ROUND(MIN(f.distance_km), 2)      AS min_distance_km,
    ROUND(MAX(f.distance_km), 2)      AS max_distance_km,
    ROUND(STDDEV_POP(f.distance_km), 2) AS sd_distance_km
FROM trip_headers t
    INNER JOIN sa_riders r           ON r.rider_id = t.rider_id
    INNER JOIN trip_fare_breakdown f ON f.trip_id  = t.trip_id
WHERE t.trip_status = 'COMPLETED'
GROUP BY r.province, t.ride_category
ORDER BY r.province, t.ride_category;
```

Figure 18 RQ2 SQL

province	ride_category	completed_trips	avg_distance_km	min_distance_km	max_distance_km	sd_distance_km
Eastern Cape	Budget (Uber Go/Bolt Go)	1202	25.28	1.54	47.98	13.18
Eastern Cape	Comfort (Uber Comfort)	1127	24.07	1.53	47.97	13.61
Eastern Cape	Large Capacity (UberXL/Bolt XL)	1200	24.52	1.54	47.92	13.64
Eastern Cape	Standard (UberX/Bolt)	1235	24.88	1.51	47.95	13.42
Gauteng	Budget (Uber Go/Bolt Go)	3727	24.66	1.50	47.99	13.41
Gauteng	Comfort (Uber Comfort)	3601	24.90	1.52	47.98	13.44
Gauteng	Large Capacity (UberXL/Bolt XL)	3734	24.76	1.50	48.00	13.37
Gauteng	Standard (UberX/Bolt)	3788	24.74	1.52	47.99	13.46
KwaZulu-Natal	Budget (Uber Go/Bolt Go)	1217	25.15	1.50	47.97	13.34
KwaZulu-Natal	Comfort (Uber Comfort)	1184	23.88	1.51	47.91	13.13
KwaZulu-Natal	Large Capacity (UberXL/Bolt XL)	1227	25.21	1.51	47.99	12.99
KwaZulu-Natal	Standard (UberX/Bolt)	1230	24.67	1.55	47.91	13.37
North West	Budget (Uber Go/Bolt Go)	1047	24.51	1.51	47.99	13.38
North West	Comfort (Uber Comfort)	1078	24.13	1.58	47.99	13.14
North West	Large Capacity (UberXL/Bolt XL)	1056	23.81	1.56	47.99	13.68
North West	Standard (UberX/Bolt)	1061	24.90	1.54	47.93	13.65
Western Cape	Budget (Uber Go/Bolt Go)	1193	24.87	1.52	47.94	13.34
Western Cape	Comfort (Uber Comfort)	1114	24.50	1.54	47.99	13.49
Western Cape	Large Capacity (UberXL/Bolt XL)	1117	24.72	1.50	47.98	13.24
Western Cape	Standard (UberX/Bolt)	1193	24.76	1.64	47.98	13.48

Figure 19 RQ2 SQL Result

6.3 RQ3 Distance, Duration and Fare by Ride Category

```sql
/*============================================================================
6.3  RQ3 | Distance, duration and fare by ride category
----------------------------------------------------------------------------
Question: How do distance, duration and fare relate for each ride category?
============================================================================ */
WITH completed AS (
    SELECT t.ride_category,
           f.distance_km,
           f.duration_minutes,
           f.total_fare_zar
    FROM trip_headers t
        INNER JOIN trip_fare_breakdown f ON f.trip_id = t.trip_id
    WHERE t.trip_status = 'COMPLETED'
)
SELECT
    ride_category,
    COUNT(*)                              AS completed_trips,
    ROUND(AVG(distance_km), 2)            AS avg_distance_km,
    ROUND(AVG(duration_minutes), 1)       AS avg_duration_min,
    ROUND(AVG(total_fare_zar), 2)         AS avg_total_fare_zar,
    ROUND( (AVG(distance_km * total_fare_zar)
             - AVG(distance_km) * AVG(total_fare_zar))
           / (STDDEV_POP(distance_km) * STDDEV_POP(total_fare_zar)), 3)
                                          AS r_distance_fare
FROM completed
GROUP BY ride_category
ORDER BY avg_total_fare_zar DESC;
```

Figure 20 RQ3 SQL

ride_category	completed_trips	avg_distance_km	avg_duration_min	avg_total_fare_zar	r_distance_fare
Standard (UberX/Bolt)	8507	24.77	76.1	464.87	0.759
Large Capacity (UberXL/Bolt XL)	8334	24.67	75.8	464.05	0.755
Budget (Uber Go/Bolt Go)	8386	24.83	76.3	462.75	0.752
Comfort (Uber Comfort)	8104	24.48	75.5	459.85	0.761

Figure 21 RQ3 SQL Result

6.4 RQ4 Tip Frequency and Average Tip

```sql
/*============================================================================
6.4  RQ4 | Tipping frequency and average tip
----------------------------------------------------------------------------
Question: How often do riders tip, and what is the average tip amount?
============================================================================ */
SELECT
    IF(GROUPING(r.province) = 1, 'ALL PROVINCES', r.province)
                                                    AS province,
    IF(GROUPING(t.ride_category) = 1, 'ALL CATEGORIES', t.ride_category)
                                                    AS ride_category,
    COUNT(*)                                        AS completed_trips,
    SUM(CASE WHEN f.tip_zar > 0 THEN 1 ELSE 0 END) AS tipped_trips,
    ROUND(100.0 * SUM(CASE WHEN f.tip_zar > 0 THEN 1 ELSE 0 END)
          / COUNT(*), 1)                            AS tip_frequency_pct,
    ROUND(AVG(CASE WHEN f.tip_zar > 0 THEN f.tip_zar END), 2)
                                                    AS avg_tip_when_tipped_zar,
    ROUND(AVG(f.tip_zar), 2)                        AS avg_tip_all_trips_zar
FROM trip_headers t
    INNER JOIN sa_riders r           ON r.rider_id = t.rider_id
    INNER JOIN trip_fare_breakdown f ON f.trip_id  = t.trip_id
WHERE t.trip_status = 'COMPLETED'
GROUP BY r.province, t.ride_category WITH ROLLUP
ORDER BY GROUPING(r.province), r.province,
         GROUPING(t.ride_category), tip_frequency_pct DESC;
```

Figure 22 RQ4 SQL

province	ride_category	completed_trips	tipped_trips	tip_frequency_pct	avg_tip_when_tipped_zar	avg_tip_all_trips_zar
Eastern Cape	Standard (UberX/Bolt)	1235	850	68.8	24.24	16.68
Eastern Cape	Budget (Uber Go/Bolt Go)	1202	807	67.1	23.01	15.45
Eastern Cape	Comfort (Uber Comfort)	1127	741	65.7	23.88	15.70
Eastern Cape	Large Capacity (UberXL/Bolt XL)	1200	783	65.3	23.54	15.36
Eastern Cape	ALL CATEGORIES	4764	3181	66.8	23.67	15.81
Gauteng	Standard (UberX/Bolt)	3788	2555	67.4	23.77	16.03
Gauteng	Budget (Uber Go/Bolt Go)	3727	2479	66.5	23.99	15.96
Gauteng	Comfort (Uber Comfort)	3601	2368	65.8	23.65	15.56
Gauteng	Large Capacity (UberXL/Bolt XL)	3734	2458	65.8	24.10	15.86
Gauteng	ALL CATEGORIES	14850	9860	66.4	23.88	15.85
KwaZulu-Natal	Large Capacity (UberXL/Bolt XL)	1227	826	67.3	22.91	15.42
KwaZulu-Natal	Comfort (Uber Comfort)	1184	781	66.0	25.04	16.52
KwaZulu-Natal	Budget (Uber Go/Bolt Go)	1217	781	64.2	24.37	15.64
KwaZulu-Natal	Standard (UberX/Bolt)	1230	790	64.2	23.84	15.31
KwaZulu-Natal	ALL CATEGORIES	4858	3178	65.4	24.02	15.72
North West	Large Capacity (UberXL/Bolt XL)	1056	745	70.5	22.88	16.14
North West	Comfort (Uber Comfort)	1078	745	69.1	23.36	16.14
North West	Budget (Uber Go/Bolt Go)	1047	701	67.0	24.24	16.23
North West	Standard (UberX/Bolt)	1061	704	66.4	24.89	16.51
North West	ALL CATEGORIES	4242	2895	68.2	23.82	16.26
Western Cape	Large Capacity (UberXL/Bolt XL)	1117	762	68.2	23.93	16.32
Western Cape	Standard (UberX/Bolt)	1193	787	66.0	23.48	15.49
Western Cape	Budget (Uber Go/Bolt Go)	1193	784	65.7	24.78	16.28
Western Cape	Comfort (Uber Comfort)	1114	730	65.5	24.22	15.87
Western Cape	ALL CATEGORIES	4617	3063	66.3	24.10	15.99
ALL PROVIN...	ALL CATEGORIES	33331	22177	66.5	23.89	15.90

Figure 23 RQ4 SQL Result

6.5 RQ5 Payment Method Adoption by Province

Research Question: What are the most frequently used payment methods among riders, and how does the uptake of different payment methods vary between the provinces of South Africa?

Technique: Common Table Expression (CTE), multi-table INNER JOIN, GROUP BY aggregation and a window function using SUM() OVER (PARTITION BY).

This query investigates the adoption of the five payment methods across the South African provinces. Only completed trips are included because the objective is to determine which payment methods were used for successfully completed transactions.

The trip_headers table is joined to sa_riders through rider_id to obtain the rider's province. It is then joined to trip_fare_breakdown through trip_id to obtain the payment method used for the trip.

A Common Table Expression named payment_counts first groups completed trips according to province and payment method and calculates the number of trips in each group. The outer query then uses SUM(trip_count) OVER (PARTITION BY province) to calculate each payment method's percentage share within its province.

```sql
/*============================================================================
6.5  RQ5 | Payment method adoption by province
----------------------------------------------------------------------------
Question: What are the most frequently used payment methods among riders,
          and how does uptake of each method vary between the provinces
          of South Africa?
============================================================================ */
WITH payment_counts AS (
    SELECT
        r.province,
        f.payment_method,
        COUNT(*) AS trip_count
    FROM trip_headers AS t

    INNER JOIN sa_riders AS r
        ON t.rider_id = r.rider_id

    INNER JOIN trip_fare_breakdown AS f
        ON t.trip_id = f.trip_id

    WHERE t.trip_status = 'COMPLETED'

    GROUP BY
        r.province,
        f.payment_method
)
SELECT
    province,
    payment_method,
    trip_count,

    ROUND(
        100.0 * trip_count /
        SUM(trip_count) OVER (PARTITION BY province),
        2
    ) AS pct_within_province

FROM payment_counts

ORDER BY
    province,
    pct_within_province DESC;
```

Figure 24 RQ5 SQL

province	payment_method	trip_count	pct_within_province
Eastern Cape	CREDIT_CARD	1073	22.52
Eastern Cape	EFT_OZOW	1053	22.10
Eastern Cape	CASH	1023	21.47
Eastern Cape	DEBIT_CARD	823	17.28
Eastern Cape	IN_APP_WALLET	792	16.62
Gauteng	CREDIT_CARD	3175	21.38
Gauteng	IN_APP_WALLET	3101	20.88
Gauteng	DEBIT_CARD	3020	20.34
Gauteng	CASH	2814	18.95
Gauteng	EFT_OZOW	2740	18.45
KwaZulu-Natal	CASH	1196	24.62
KwaZulu-Natal	IN_APP_WALLET	961	19.78
KwaZulu-Natal	DEBIT_CARD	945	19.45
KwaZulu-Natal	EFT_OZOW	916	18.86
KwaZulu-Natal	CREDIT_CARD	840	17.29
North West	DEBIT_CARD	898	21.17
North West	EFT_OZOW	893	21.05
North West	CASH	864	20.37
North West	IN_APP_WALLET	811	19.12
North West	CREDIT_CARD	776	18.29
Western Cape	CASH	1197	25.93
Western Cape	CREDIT_CARD	978	21.18
Western Cape	EFT_OZOW	859	18.61
Western Cape	DEBIT_CARD	836	18.11
Western Cape	IN_APP_WALLET	747	16.18

Figure 25 RQ5 SQL Result

6.6 RQ6 Preferred vs. Actual Payment Method

Research Question: To what extent is the rider's actual mode of payment different from their preferred mode of payment?

Technique: Multi-table INNER JOIN, GROUP BY, conditional aggregation using CASE and scalar subqueries.

Three related queries are used to answer this research question. The first creates a cross-tabulation between the rider's stored preferred payment method and the actual payment method used on completed trips.

6.6.1 Preferred vs. Actual Payment Cross-Tabulation

```sql
-- ------------------------------------------------------------------------
-- 6.6.1 Number of completed trips for every preferred/actual combination.
-- ------------------------------------------------------------------------
SELECT
    r.preferred_payment,
    f.payment_method AS actual_payment_method,
    COUNT(*) AS trip_count

FROM trip_headers AS t

INNER JOIN sa_riders AS r
    ON t.rider_id = r.rider_id

INNER JOIN trip_fare_breakdown AS f
    ON t.trip_id = f.trip_id

WHERE t.trip_status = 'COMPLETED'

GROUP BY
    r.preferred_payment,
    f.payment_method

ORDER BY
    r.preferred_payment,
    trip_count DESC;
```

Figure 26 RQ6 - 6.6.1 SQL

	preferred_payment	actual_payment_method	trip_count
▶	CASH	CASH	7094
	CREDIT_CARD	CREDIT_CARD	6842
	DEBIT_CARD	DEBIT_CARD	6522
	EFT_OZOW	EFT_OZOW	6461
	IN_APP_WALLET	IN_APP_WALLET	6412

Figure 27 RQ6 - 6.6.1 SQL Result

The second query uses CASE expressions to count matching and non-matching payment choices and calculate the overall match and mismatch percentages.

6.6.2 Overall Match and Mismatch Rate

```sql
SELECT
    COUNT(*) AS completed_trips,

    SUM(
        CASE
            WHEN r.preferred_payment = f.payment_method
            THEN 1
            ELSE 0
        END
    ) AS matching_trips,

    SUM(
        CASE
            WHEN r.preferred_payment <> f.payment_method
            THEN 1
            ELSE 0
        END
    ) AS non_matching_trips,

    ROUND(
        100.0 *
        SUM(
            CASE
                WHEN r.preferred_payment = f.payment_method
                THEN 1
                ELSE 0
            END
        ) / COUNT(*),
        2
    ) AS match_percentage,

    ROUND(
        100.0 *
        SUM(
            CASE
                WHEN r.preferred_payment <> f.payment_method
                THEN 1
                ELSE 0
            END
        ) / COUNT(*),
        2
    ) AS mismatch_percentage

FROM trip_headers AS t

INNER JOIN sa_riders AS r
    ON t.rider_id = r.rider_id

INNER JOIN trip_fare_breakdown AS f
    ON t.trip_id = f.trip_id

WHERE t.trip_status = 'COMPLETED';
```

Figure 28 RQ6 - 6.6.2 SQL

	completed_trips	matching_trips	non_matching_trips	match_percentage	mismatch_percentage
▶	33331	33331	0	100.00	0.00

Figure 29 RQ6 - 6.6.2 SQL Result

A third query demonstrates the use of scalar subqueries by independently calculating the percentage of completed trips where the preferred payment method matches the actual payment method.

6.6.3 Match Percentage Using Subqueries

```sql
-- ------------------------------------------------------------------------
-- 6.6.3 Match percentage (subquery)
-- ------------------------------------------------------------------------
SELECT
    ROUND(
        100.0 *
        (
            SELECT COUNT(*)
            FROM trip_headers AS t2

            INNER JOIN sa_riders AS r2
                ON t2.rider_id = r2.rider_id

            INNER JOIN trip_fare_breakdown AS f2
                ON t2.trip_id = f2.trip_id

            WHERE t2.trip_status = 'COMPLETED'
              AND r2.preferred_payment = f2.payment_method
        )
        /
        (
            SELECT COUNT(*)
            FROM trip_headers AS t3
            WHERE t3.trip_status = 'COMPLETED'
        ),
        2
    ) AS preferred_actual_match_percentage;
```

Figure 30 RQ6 - 6.6.3 SQL

	preferred_actual_match_percentage
▶	100.00

Figure 31 RQ6 - 6.6.3 SQL Result

The results produced 33,331 matching completed trips and zero non-matching trips, resulting in a 100.00% match rate. This result is interpreted carefully in Section 7.6 because the dataset contains no observable payment-switching behaviour.

6.7 RQ7 Payment Method and Trip Cancellations

Research Question: Is there a relationship between the payment method and the likelihood of a rider-initiated or driver-initiated cancellation of a trip?

Technique: INNER JOIN, conditional aggregation using CASE, GROUP BY, HAVING and percentage calculations.

This query includes all trip statuses because cancellations are the outcome being investigated. It groups trips by payment method and uses CASE expressions to count completed trips, rider cancellations and driver cancellations. It also calculates separate rider, driver and total cancellation percentages.

The HAVING COUNT(*) >= 100 condition ensures that only payment-method groups with a sufficient number of observations are retained. All five payment methods satisfy this condition in the supplied dataset.

```sql
/*============================================================================
6.7  RQ7 | Payment method and trip cancellations
----------------------------------------------------------------------------
Question : Is the payment method related to how often trips are cancelled?
============================================================================ */
SELECT f.payment_method,
    COUNT(*) AS total_trips,

    SUM(
        CASE
            WHEN t.trip_status = 'COMPLETED'
            THEN 1
            ELSE 0
        END
    ) AS completed_trips,

    SUM(
        CASE
            WHEN t.trip_status = 'CANCELLED_BY_RIDER'
            THEN 1
            ELSE 0
        END
    ) AS cancelled_by_rider,

    SUM(
        CASE
            WHEN t.trip_status = 'CANCELLED_BY_DRIVER'
            THEN 1
            ELSE 0
        END
    ) AS cancelled_by_driver,
```

Figure 32 RQ7 Part1 SQL

```
        ROUND(
            100.0 *
            SUM(
                CASE
                    WHEN t.trip_status = 'CANCELLED_BY_RIDER'
                    THEN 1
                    ELSE 0
                END
            ) / COUNT(*),
            2
        ) AS rider_cancellation_pct,

        ROUND(
            100.0 *
            SUM(
                CASE
                    WHEN t.trip_status = 'CANCELLED_BY_DRIVER'
                    THEN 1
                    ELSE 0
                END
            ) / COUNT(*),
            2
        ) AS driver_cancellation_pct,

```

Figure 33 RQ7 Part2 SQL

```
348         ROUND(
349             100.0 *
350             SUM(
351                 CASE
352                     WHEN t.trip_status IN (
353                         'CANCELLED_BY_RIDER',
354                         'CANCELLED_BY_DRIVER'
355                     )
356                     THEN 1
357                     ELSE 0
358                 END
359             ) / COUNT(*),
360             2
361         ) AS total_cancellation_pct
362
363     FROM trip_headers AS t
364
365     INNER JOIN trip_fare_breakdown AS f
366         ON t.trip_id = f.trip_id
367
368     GROUP BY
369         f.payment_method
370
371     HAVING COUNT(*) >= 100
372
373     ORDER BY
374         total_cancellation_pct DESC;
```

Figure 34 RQ7 Part3 SQL

payment_method	total_trips	completed_trips	cancelled_by_rider	cancelled_by_driver	rider_cancellation_pct	driver_cancellation_pct	total_cancellation_pct
CREDIT_CARD	10353	6842	1792	1719	17.31	16.60	33.91
DEBIT_CARD	9863	6522	1660	1681	16.83	17.04	33.87
EFT_OZOW	9704	6461	1622	1621	16.71	16.70	33.42
IN_APP_WALLET	9575	6412	1624	1539	16.96	16.07	33.03
CASH	10505	7094	1719	1692	16.36	16.11	32.47

Figure 35 RQ7 SQL Result

The results show very similar cancellation rates across all payment channels. Credit Card has the highest overall cancellation rate at 33.91%, while Cash has the lowest at 32.47%, a difference of only 1.44 percentage points.

6.8 RQ8 Surge Pricing and Rider Spending

Research Question: How does surge pricing affect rider spending across different ride categories and payment methods?

Technique: Common Table Expressions, CASE-based surge bucketing, INNER JOIN, multi-column GROUP BY aggregation, AVG() and SUM().

Two queries are used to answer RQ8. Only completed trips are included because the analysis investigates actual rider expenditure. The surge multiplier is grouped into four bands: No Surge (1.00), Low Surge (>1.00 to 1.50), Medium Surge (>1.50 to 2.00) and High Surge (>2.00).

6.8.1 Surge Pricing by Ride Category

The first query groups completed trips by surge band and ride category and calculates the number of trips, average fare and total spending for each combination.

```sql
-- ------------------------------------------------------------------------
-- 6.8.1  RQ8 | Surge pricing by ride category
-- ------------------------------------------------------------------------
WITH surge_data AS (
    SELECT
        CASE
            WHEN f.surge_multiplier = 1.00
                THEN '1. No Surge'

            WHEN f.surge_multiplier > 1.00
                 AND f.surge_multiplier <= 1.50
                THEN '2. Low Surge'

            WHEN f.surge_multiplier > 1.50
                 AND f.surge_multiplier <= 2.00
                THEN '3. Medium Surge'

            ELSE '4. High Surge'
        END AS surge_band,

        t.ride_category,
        f.total_fare_zar

    FROM trip_headers AS t

    INNER JOIN trip_fare_breakdown AS f
        ON t.trip_id = f.trip_id

    WHERE t.trip_status = 'COMPLETED'
)
SELECT
    surge_band,
    ride_category,
    COUNT(*) AS trip_count,
    ROUND(AVG(total_fare_zar), 2) AS avg_fare_zar,
    ROUND(SUM(total_fare_zar), 2) AS total_spend_zar

FROM surge_data

GROUP BY
    surge_band,
    ride_category

ORDER BY
    ride_category,
    surge_band;
```

Figure 36 RQ8 - 6.8.1 SQL

surge_band	ride_category	trip_count	avg_fare_zar	total_spend_zar
1. No Surge	Budget (Uber Go/Bolt Go)	5663	369.47	2092314.24
2. Low Surge	Budget (Uber Go/Bolt Go)	654	535.80	350413.14
3. Medium Surge	Budget (Uber Go/Bolt Go)	1215	628.55	763686.66
4. High Surge	Budget (Uber Go/Bolt Go)	854	789.43	674173.02
1. No Surge	Comfort (Uber Comfort)	5419	361.36	1958204.58
2. Low Surge	Comfort (Uber Comfort)	600	513.43	308058.74
3. Medium Surge	Comfort (Uber Comfort)	1238	634.08	784987.34
4. High Surge	Comfort (Uber Comfort)	847	797.39	675388.09
1. No Surge	Large Capacity (UberXL/Bolt XL)	5564	365.88	2035763.05
2. Low Surge	Large Capacity (UberXL/Bolt XL)	636	530.20	337209.15
3. Medium Surge	Large Capacity (UberXL/Bolt XL)	1253	637.34	798588.50
4. High Surge	Large Capacity (UberXL/Bolt XL)	881	789.86	695865.63
1. No Surge	Standard (UberX/Bolt)	5747	368.65	2118617.76
2. Low Surge	Standard (UberX/Bolt)	640	506.30	324030.21
3. Medium Surge	Standard (UberX/Bolt)	1271	665.23	845513.68
4. High Surge	Standard (UberX/Bolt)	849	785.06	666517.15

Figure 37 RQ8 - 6.8.1 SQL Result

6.8.2 Surge Pricing by Payment Method

The second query follows the same structure but groups the completed trips according to payment method instead of ride category.

```sql
-- ------------------------------------------------------------------------
-- 6.8.2  RQ8 | Surge pricing by payment method
-- ------------------------------------------------------------------------
WITH surge_data AS (
    SELECT
        CASE
            WHEN f.surge_multiplier = 1.00
                THEN '1. No Surge'

            WHEN f.surge_multiplier > 1.00
                 AND f.surge_multiplier <= 1.50
                THEN '2. Low Surge'

            WHEN f.surge_multiplier > 1.50
                 AND f.surge_multiplier <= 2.00
                THEN '3. Medium Surge'

            ELSE '4. High Surge'
        END AS surge_band,

        f.payment_method,
        f.total_fare_zar

    FROM trip_headers AS t

    INNER JOIN trip_fare_breakdown AS f
        ON t.trip_id = f.trip_id

    WHERE t.trip_status = 'COMPLETED'
)
SELECT
    surge_band,
    payment_method,
    COUNT(*) AS trip_count,
    ROUND(AVG(total_fare_zar), 2) AS avg_fare_zar,
    ROUND(SUM(total_fare_zar), 2) AS total_spend_zar

FROM surge_data

GROUP BY
    surge_band,
    payment_method

ORDER BY
    payment_method,
    surge_band;
```

Figure 38 RQ8 - 6.8.2 SQL

surge_band	payment_method	trip_count	avg_fare_zar	total_spend_zar
1. No Surge	CASH	4786	365.76	1750508.08
2. Low Surge	CASH	552	519.93	287001.37
3. Medium Surge	CASH	1041	632.04	657952.21
4. High Surge	CASH	715	777.92	556213.44
1. No Surge	CREDIT_CARD	4625	364.94	1687854.30
2. Low Surge	CREDIT_CARD	528	534.00	281953.66
3. Medium Surge	CREDIT_CARD	1012	670.68	678729.23
4. High Surge	CREDIT_CARD	677	782.35	529648.24
1. No Surge	DEBIT_CARD	4335	368.58	1597782.09
2. Low Surge	DEBIT_CARD	489	517.34	252977.78
3. Medium Surge	DEBIT_CARD	996	630.85	628325.33
4. High Surge	DEBIT_CARD	702	804.64	564858.90
1. No Surge	EFT_OZOW	4384	370.80	1625583.81
2. Low Surge	EFT_OZOW	461	526.32	242633.40
3. Medium Surge	EFT_OZOW	943	631.73	595722.33
4. High Surge	EFT_OZOW	673	788.37	530573.34
1. No Surge	IN_APP_WALLET	4263	361.99	1543171.35
2. Low Surge	IN_APP_WALLET	500	510.29	255145.03
3. Medium Surge	IN_APP_WALLET	985	641.67	632047.08
4. High Surge	IN_APP_WALLET	664	799.17	530649.97

Figure 39 RQ8 - 6.8.2 SQL Result

Both queries show the same strong upward pattern as the surge band increases. The detailed interpretation is presented in Section 7.8.

6.9 Supporting Query - Satisfaction vs. Payment Method

This supporting query aims to determine whether the satisfaction of the riders is affected by the method of payment that was chosen during a particular trip. The query uses tables: trip_headers, trip_fare_breakdown, and trip_reviews. The trip_headers table is connected to trip_fare_breakdown through the trip_id in order to get the method of payment.

The connection is made via the LEFT OUTER JOIN that connects trip_reviews to trip_headers. The use of the LEFT OUTER JOIN is necessary as not all completed trips have reviews. If we would use the INNER JOIN, we could eliminate some trips that do not have a corresponding record in the trip_reviews table.

The result will be grouped by the payment method, and for each payment method, the number of completed trips, the number of reviewed trips, and the average rider rating will be counted.

```sql
/*============================================================================
 6.9  RQ9 | Rider satisfaction by payment method
----------------------------------------------------------------------------
Question : Does rider satisfaction differ by payment method?
============================================================================ */
SELECT
    f.payment_method,
    COUNT(t.trip_id) AS completed_trips,
    COUNT(rv.review_id) AS reviewed_trips,
    ROUND(AVG(rv.rider_rating_of_driver), 2) AS avg_rider_rating_of_driver
FROM trip_headers t
JOIN trip_fare_breakdown f
    ON t.trip_id = f.trip_id
LEFT OUTER JOIN trip_reviews rv
    ON t.trip_id = rv.trip_id
WHERE t.trip_status = 'COMPLETED'
GROUP BY f.payment_method
ORDER BY avg_rider_rating_of_driver DESC;
```

Figure 40 6.9 Support Query SQL

payment_method	completed_trips	reviewed_trips	avg_rider_rating_of_driver
DEBIT_CARD	6522	4816	3.66
EFT_OZOW	6461	4858	3.66
CREDIT_CARD	6842	5139	3.65
CASH	7094	5347	3.64
IN_APP_WALLET	6412	4895	3.62

Figure 41 6.9 Support Query SQL Result

6.10 Supporting Query - High-Value Riders

This query retrieves those riders whose average spent fare per completed trip is greater than the average fare of completed trips on the whole platform. The join operations use rider_id between trip_headers and sa_riders and trip_id between trip_headers and trip_fare_breakdown. Only completed trips are taken into consideration in order to exclude cancellation fees which have no monetary value.

The query groups the data by rider and province and computes the average fare for each rider along with his number of completed trips. A subquery finds out the average fare of all completed trips on the platform. Finally, the HAVING clause filters the results such that only those riders whose average spent fare is higher than the computed average are selected.

```sql
/* ============================================================================
6.10  RQ10 | High-value riders
----------------------------------------------------------------------------
   Question : Which riders are "high value"?
============================================================================ */
SELECT  r.rider_id, r.province,
        COUNT(*)                        AS completed_trips,
        ROUND(AVG(f.total_fare_zar), 2) AS avg_fare_zar
FROM    trip_headers t
JOIN    sa_riders r ON t.rider_id = r.rider_id
JOIN    trip_fare_breakdown f ON t.trip_id = f.trip_id
WHERE   t.trip_status = 'COMPLETED'
GROUP BY r.rider_id, r.province
HAVING  AVG(f.total_fare_zar) > (
            SELECT AVG(f2.total_fare_zar)
            FROM   trip_headers t2
            JOIN   trip_fare_breakdown f2 ON t2.trip_id = f2.trip_id
            WHERE  t2.trip_status = 'COMPLETED'
        )
ORDER BY avg_fare_zar DESC;
```

Figure 42 6.10 Support Query SQL

rider_id	province	completed_trips	avg_fare_zar
RIDER_000216	Gauteng	27	666.64
RIDER_000672	Western Cape	19	650.21
RIDER_000536	Eastern Cape	29	616.02
RIDER_001125	Gauteng	19	614.50
RIDER_000472	Gauteng	20	610.86
RIDER_000852	Western Cape	21	608.19
RIDER_000521	KwaZulu-Natal	20	607.38
RIDER_001028	Gauteng	23	607.31
RIDER_000825	Western Cape	27	601.94
RIDER_000246	KwaZulu-Natal	21	599.43
RIDER_000561	Eastern Cape	23	598.51
RIDER_000078	KwaZulu-Natal	15	597.72
RIDER_001067	Gauteng	19	597.41
RIDER_000442	Gauteng	28	596.79
RIDER_000960	KwaZulu-Natal	24	596.29
RIDER_000422	Gauteng	26	592.22
RIDER_000634	North West	26	592.09

Figure 43 6.10 Support Query SQL Result

6.11 Supporting Query - Transaction Example

Purpose: To demonstrate an explicit database transaction in MySQL by making a controlled correction to a fare record, checking the change, and then reversing it with ROLLBACK. This shows how related financial values can be updated safely as one atomic operation.

Technique: START TRANSACTION, user variables, SELECT verification, UPDATE, and ROLLBACK. COMMIT is discussed as the alternative when a verified correction must be saved permanently.

A transaction is useful when more than one value must remain consistent. In this example, an incorrectly captured toll is removed from a trip. The toll amount and the total fare are linked financially, so both values must be corrected together. If the correction is wrong, ROLLBACK restores the original record. If the correction has been checked and is correct, COMMIT would permanently save it.

6.11.1 MySQL Transaction Demonstration

The query first selects an existing trip that has a positive toll and stores its trip identifier and original toll amount in MySQL variables. This avoids using an invented trip ID. The selected record is displayed before the change, updated inside the transaction, displayed again, and finally restored with ROLLBACK.

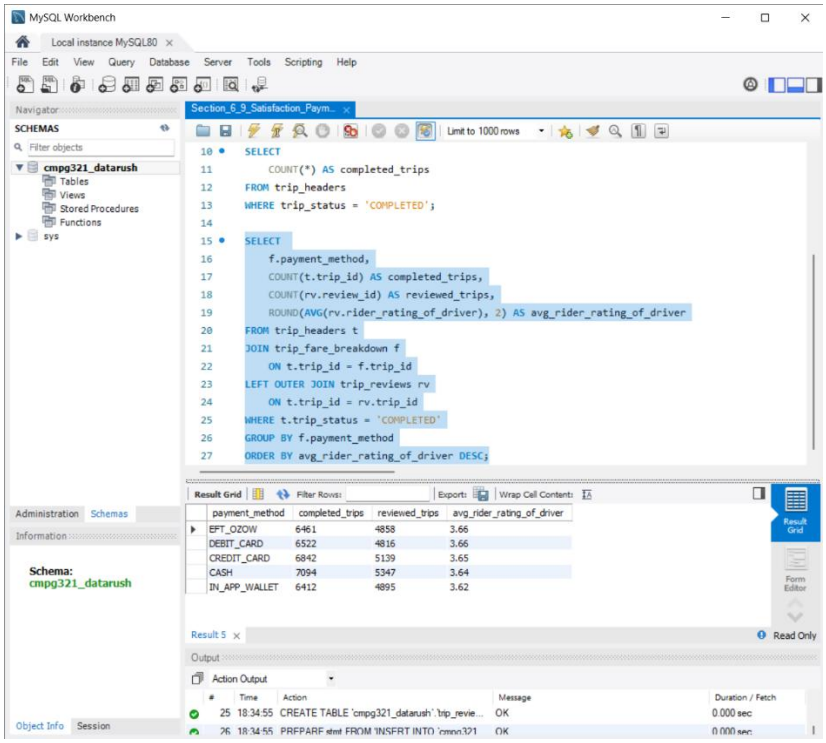


Figure 44 6.11.1 Support Query SQL Result

```sql
/* ============================================================================
6.11 Supporting Query: Transaction Example
============================================================================ */
SET @trip_to_correct = (
    SELECT trip_id
    FROM trip_fare_breakdown
    WHERE tolls_zar > 0
      AND total_fare_zar >= tolls_zar
    ORDER BY trip_id
    LIMIT 1
);

SET @original_toll = (
    SELECT tolls_zar
    FROM trip_fare_breakdown
    WHERE trip_id = @trip_to_correct
);

START TRANSACTION;

SELECT
    trip_id,
    tolls_zar,
    total_fare_zar
FROM trip_fare_breakdown
WHERE trip_id = @trip_to_correct;

UPDATE trip_fare_breakdown
SET
    tolls_zar = 0.00,
    total_fare_zar = total_fare_zar - @original_toll
WHERE trip_id = @trip_to_correct;

SELECT
    trip_id,
    tolls_zar,
    total_fare_zar
FROM trip_fare_breakdown
WHERE trip_id = @trip_to_correct;

ROLLBACK;

SELECT
    trip_id,
    tolls_zar,
    total_fare_zar
FROM trip_fare_breakdown
WHERE trip_id = @trip_to_correct;
```

Figure 45 6.11.1 Support Query SQL

6.11.2 Transaction Explanation

START TRANSACTION: begins the transaction and ensures that the following change can either be saved as a unit or reversed.

SELECT before the UPDATE: shows the original toll and total fare so that the starting values are known.

UPDATE: sets the toll to zero and subtracts the same toll amount from total_fare_zar. This keeps the financial values internally consistent.

SELECT after the UPDATE: check the temporary result before deciding whether it should be saved.

ROLLBACK: cancels the change and restores the original values. This is the safest option for a classroom demonstration because it does not permanently alter the supplied dataset.

COMMIT: would be used instead of ROLLBACK if the correction had been verified and needed to be stored permanently.

The transaction demonstrates the ACID principle of atomicity because the related fare changes are treated as one logical unit. The database does not need to keep a partially corrected record. The change can either be committed as a complete correction or rolled back to the original state.

The transaction sequence is illustrated below. It shows the intended before-update-rollback flow without claiming a separate Workbench execution result for Section 6.11.

7. Analysis and Reporting

This section analyses and interprets the results that were obtained from the SQL queries in Section 6. The findings are used to identify patterns in rider spending payment behaviour, highlight important insights and provide business recommendations, while considering the limitations of the available dataset.

7.1 RQ1 -- Riders' Provincial Spending Patterns

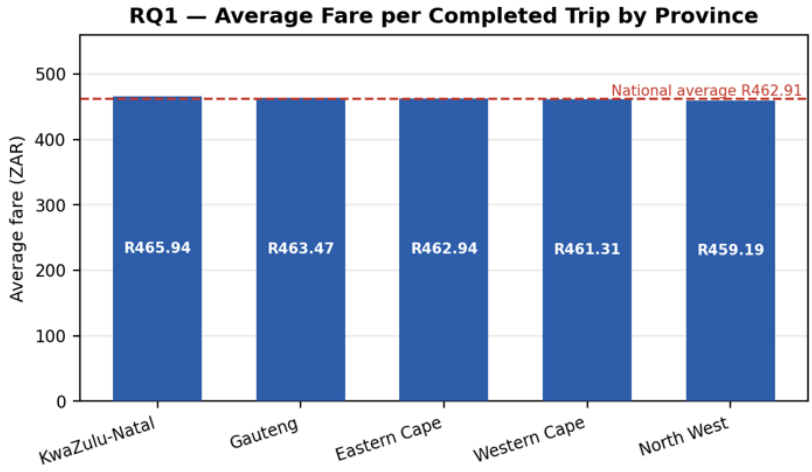


Figure 46 7.1 Riders Provincial Spending Patterns Chart

Province	Avg. Fare per Completed Trip	Difference vs. National Avg.	Rank
KwaZulu-Natal	R465.94	+R3.03	1
Gauteng	R463.47	+R0.56	2
Eastern Cape	R462.94	+R0.03	3
Western Cape	R461.31	−R1.60	4
North West	R459.19	−R3.72	5

The average amount spent per completed trip is basically the same across all the provinces. KwaZulu-Natal has the highest average at R465.94, while North West has the lowest at R459.19. This is only a difference of R6.75. No province is also more than R3.72 away from the national average of R462.91. Since individual completed trips can cost almost R2,000, a difference of only a few rand between the provincial averages is too small to say that riders from one province actually spend more than riders from another.

There are two important things to keep in mind when looking at these results. First, the province used here is the rider's home province from `sa_riders.province`, and not the province where the trip actually happened. This was done because RQ1 focuses on how much riders spend. If we wanted to look at where the trips happened instead, `trip_headers` would need to be joined to `pricing_surge_zones` using `zone_id`.

Second, Query 6.1 uses `RANK()` to rank the provinces, but the differences between them are very small. Because of this, the rankings could easily change if a different sample of trips was used. So, even though KwaZulu-Natal is technically ranked first, it would not really be correct to say that it is a significantly higher-spending province.

The main difference between the provinces is actually the number of trips completed. Gauteng alone has 14,850 completed trips, which makes up 44.6% of all completed trips, while its average fare is still very close to the national average. This means that differences in revenue between provinces mostly come from the number of trips being completed, rather than customers paying more per trip.

Since about a third of all trip requests end up being cancelled, as shown in Section 7.7, improving driver availability and getting more trips completed would probably have a bigger impact for Uber and Bolt than changing prices based on the province.

7.2 RQ2 Trip Distance Distribution

Trip distance follows basically the same pattern as spending. Across all 20 provinces × ride-category groups in Query 6.2, the average completed-trip distance is between 23.8 km and 25.3 km. This is only a difference of 1.5 km. No province consistently has shorter or longer trips, and none of the ride categories are clearly used for shorter or longer trips either. At category level, the averages are all between 24.5 km and 24.8 km, as shown in Section 7.3.

This also rules out the simple idea that riders use Budget for shorter trips and Large Capacity for longer trips. In this dataset, the category chosen does not seem to have anything to do with distance. This means that category choice would more likely depend on other things, such as price, group size or vehicle preference, but the dataset does not contain enough information to test these factors.

The fact that the distances are so similar is also important. In real ride-hailing data, you would normally expect some differences between regions. For example, riders in larger and more spread-out cities may travel further than riders in smaller towns. In this dataset, however, the average distance is almost the same whether the rider is from Gauteng or North West. Together with the very similar spending found in RQ1 and the 100% match between preferred and actual payment method in RQ6 (Section 7.6), this suggests that trip distance may have been generated separately from province and ride category.

Because of this, the distance results should mainly be seen as a feature of this specific dataset, and not as strong evidence of how South African riders actually travel.

7.3 RQ3 Distance, Duration and Fare

Distance is the strongest factor affecting how much a rider pays. In every ride category, the Pearson correlation between distance and total fare in Query 6.3 is between r = 0.752 and r = 0.761. If these values are squared, distance accounts for approximately 57–58% of the variation in fare within each category.

The rest of the difference in fares comes from other factors that are not fully shown by a single correlation. One of the main factors is the surge multiplier, which by itself can more than double the average fare, as shown in Section 7.8. Other things such as tolls, tips and different zone tariffs also affect the final amount.

Trip duration does not add much when comparing the different categories. The average completed trip takes around 76 minutes in every category, while the average distance remains between 24.5 km and 24.8 km. Duration is still included in the fare through the per-minute rate in pricing_surge_zones, but because the average duration is almost the same for every category, it does not explain the differences between them.

The ride category itself also seems to make very little difference to the fare. Average fares range from R459.85 for Comfort to R464.87 for Standard, which is only about a R5 difference, or around 1%. Large Capacity has an average of R464.05 and Budget averages R462.75.

This result is unusual because Comfort is normally seen as a more premium option, but in this dataset, it actually has the lowest average fare. Budget is also not cheaper than Standard. On real ride-hailing platforms, these categories normally have different pricing. The fact that there is almost no category difference here suggests that fares in this dataset were mainly calculated using distance, duration and surge, without a separate multiplier for the ride category.

For the focus area, this means that rider spending is mainly affected by how far they travel and whether surge pricing applies, rather than which ride category they choose. Because the categories are priced almost the same in this dataset, it is not really possible to compare different category pricing strategies using these results.

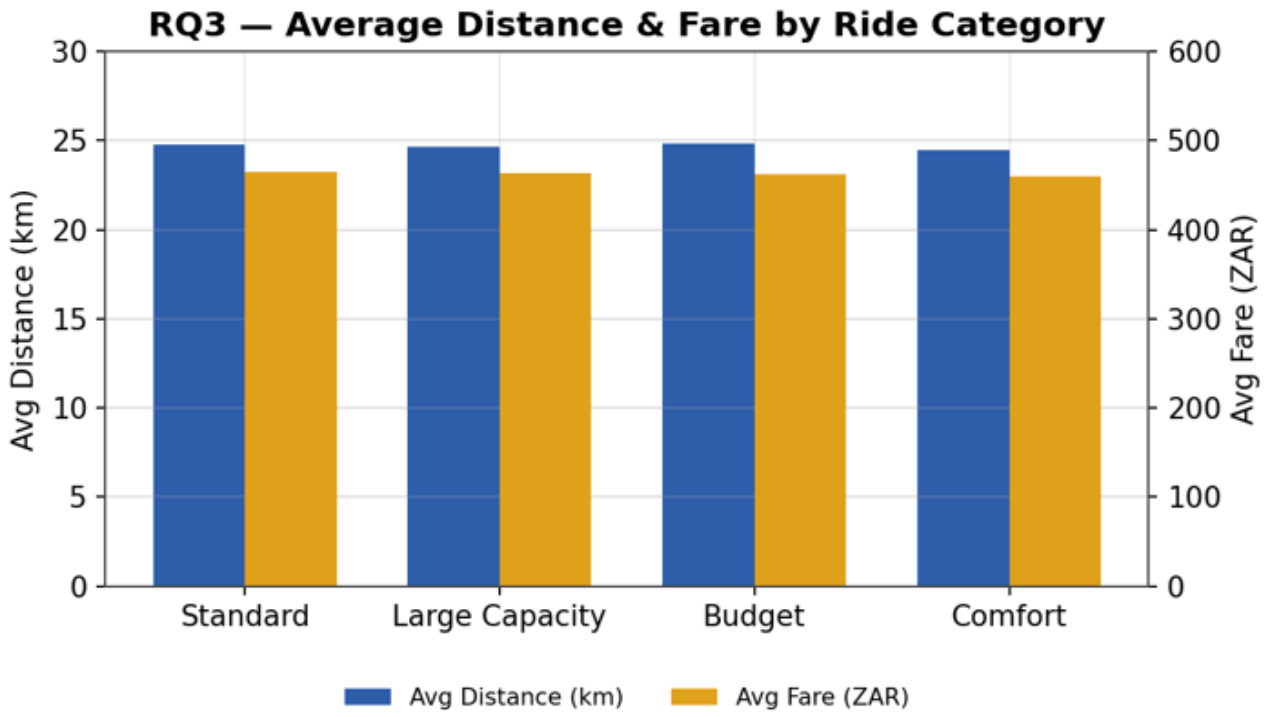


Figure 47 Average Distance & Fare by Ride Cat. Chart

7.4 RQ4 Tip Frequency

Tipping is fairly common in the dataset, with 66.4% of completed trips receiving a tip. The province subtotal rows created by the ROLLUP in Query 6.4 range from 65.4% in KwaZulu-Natal to 68.2% in North West. This is a difference of only 2.8 percentage points.

The more detailed province × category results vary a bit more. These range from 64.2% for Standard and Budget trips in KwaZulu-Natal to 70.5% for Large Capacity trips in North West.

Most of this larger difference can probably be explained by the smaller group sizes. Outside Gauteng, each province × category group has around 1,150 completed trips on average. With groups of this size, normal random variation can move the tipping percentage by a few points. Because of this, the detailed rows should not really be treated as a ranking of which groups tip the most.

The province totals are more useful because they are based on thousands of trips. North West does show a slightly higher tipping rate, but the difference is still less than three percentage points, so it is fairly small in practical terms.

The actual tip amount varies even less. When riders do leave a tip, the average is around R22 to R25 across all the groups, with a national average of R23.90. This is very close to what would be expected from the way the tip field is set up. Phase 1 showed that tip_zar only contains four possible amounts: R10, R15, R20 and R50. If these four values were selected equally often, the average would be R23.75.

This means that riders are basically choosing from a fixed set of tip values. Because of this, the dataset tells us more about whether someone chooses to tip than how generous they are when they do.

The main business point is therefore the frequency of tipping. Across the 33,331 completed trips, the average tip works out to R15.90 per trip, which is roughly R530,000 in total tips. At the same time, around one third of completed trips did not give the driver any extra money.

Since most of the difference comes from whether a rider tips rather than the amount they choose, a national approach, such as showing a default tip suggestion after every trip, would make more sense than creating different tipping prompts for each province.

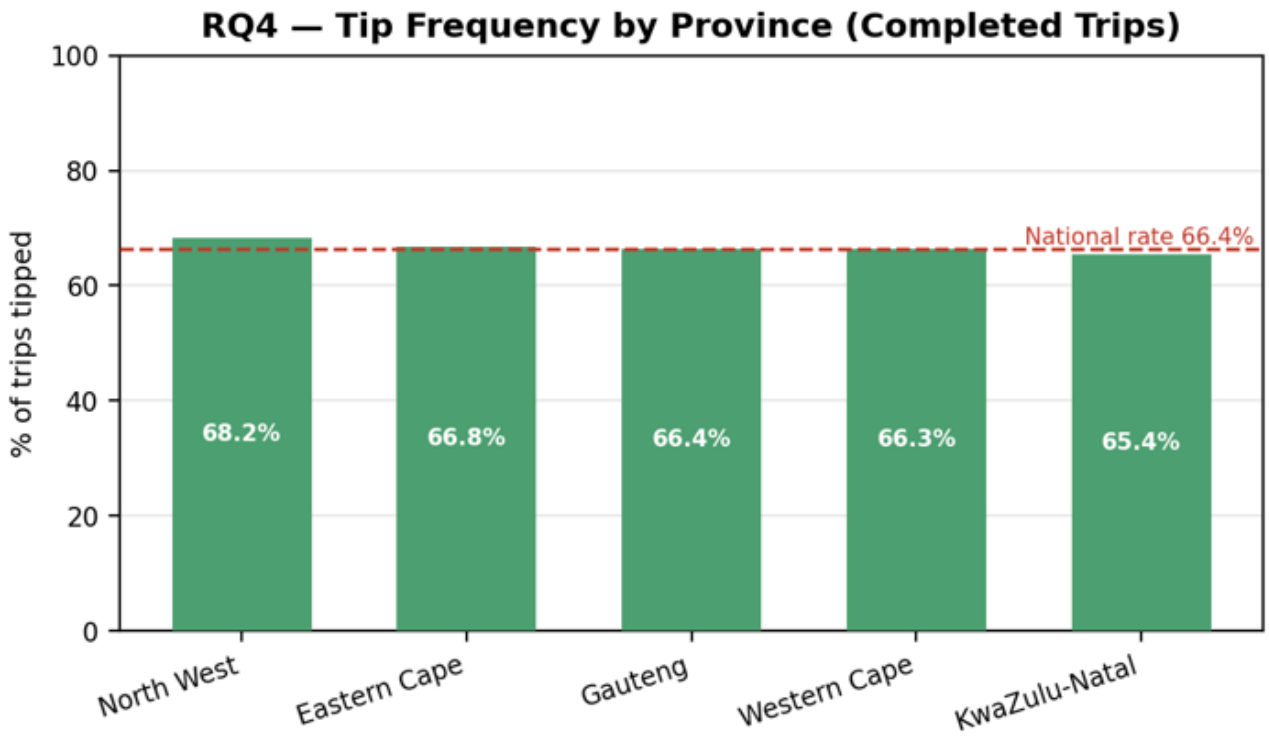


Figure 48 Tip Frequency by Province Chart

7.5 RQ5 Payment Method Adoption

Research Question: What are the most frequently used payment methods among riders, and how does the uptake of different payment methods vary between the provinces of South Africa?

The SQL results show that all five payment methods are actively used across the provinces, but the relative popularity of each payment channel differs geographically. In the Eastern Cape, Credit Card has the largest share at 22.52%, followed closely by EFT/Ozow at 22.10%. Gauteng also has Credit Card as its most frequently used payment method at 21.38%, although payment usage is relatively evenly distributed across all five methods.

KwaZulu-Natal and the Western Cape show a stronger preference for Cash. Cash accounts for 24.62% of completed trips in KwaZulu-Natal and 25.93% in the Western Cape. The Western Cape demonstrates the greatest provincial difference between the most and least used methods, with Cash at 25.93% compared with In-App Wallet at 16.18%. In the North West, Debit Card has the highest adoption at 21.17%, only slightly ahead of EFT/Ozow at 21.05%.

Leading Payment Method by Province

Province	Most Used Method	Share	Least Used Method	Share
Eastern Cape	Credit Card	22.52%	In-App Wallet	16.62%
Gauteng	Credit Card	21.38%	EFT/Ozow	18.45%
KwaZulu-Natal	Cash	24.62%	Credit Card	17.29%
North West	Debit Card	21.17%	Credit Card	18.29%
Western Cape	Cash	25.93%	In-App Wallet	16.18%

Across all 33,331 completed trips, Cash is the most frequently used method with 7,094 trips (21.28%), followed by Credit Card with 6,842 (20.53%), Debit Card with 6,522 (19.57%), EFT/Ozow with 6,461 (19.38%) and In-App Wallet with 6,412 (19.24%). These totals show that no payment method dominates nationally; the difference between the highest and lowest national shares is only about two percentage points.

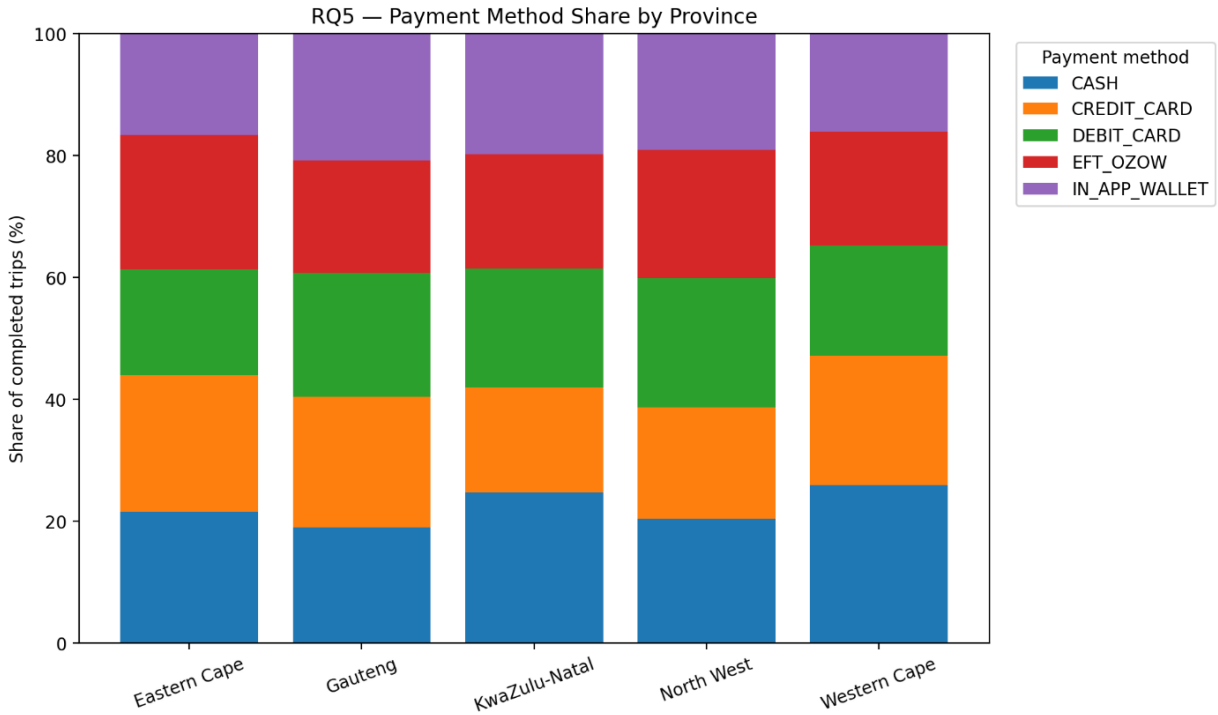


Figure 49 Payment Method Share by Province Chart

Key Insight and Business Implication

The results indicate that payment preferences differ geographically, but the national payment market is highly diversified. Uber/Bolt should therefore maintain support for all five payment channels rather than prioritising only one.

The provincial differences also create opportunities for targeted strategies. For example, if reducing the operational costs or risks associated with cash transactions is a business objective, digital-payment incentives could be investigated specifically in the Western Cape and KwaZulu-Natal, where Cash has the strongest relative adoption.

Conclusion: No single payment method dominates nationally, but meaningful provincial differences exist. Credit Card leads in the Eastern Cape and Gauteng, Cash leads in KwaZulu-Natal and the Western Cape, and Debit Card narrowly leads in the North West.

7.6 RQ6 Preferred vs. Actual Payment Method

Research Question: To what extent is the rider's actual mode of payment different from their preferred mode of payment?

The comparison between the riders stored preferred payment method and the actual payment method recorded for completed trips produces an unusually clear result. Every payment preference matches the actual payment channel used.

Cash preference corresponds to Cash for 7,094 trips, Credit Card to Credit Card for 6,842 trips, Debit Card to Debit Card for 6,522 trips, EFT/Ozow to EFT/Ozow for 6,461 trips, and In-App Wallet to In-App Wallet for 6,412 trips. No cross-payment combinations were returned.

Preferred vs. Actual Payment Method

Preferred Payment	Actual Payment	Completed Trips
Cash	Cash	7,094
Credit Card	Credit Card	6,842
Debit Card	Debit Card	6,522
EFT/Ozow	EFT/Ozow	6,461
In-App Wallet	In-App Wallet	6,412

The summary query confirms that all 33,331 completed trips matched, with zero non-matching trips. This produces a 100.00% match rate and a 0.00% mismatch rate. The separate subquery used to calculate the overall match rate independently produced the same 100.00% result.

Payment Preference Match Summary

Measure	Result
Completed trips	33,331
Matching preferred/actual trips	33,331

Measure	Result
Non-matching trips	0
Match rate	100.00%
Mismatch rate	0.00%

Key Insight and Data Limitation

At face value, the result suggests that riders always use their preferred payment method. However, a perfect match across more than 33,000 transactions should be interpreted cautiously.

The dataset shows no observable payment-switching behaviour at all. It therefore cannot be used to investigate situations such as a rider preferring Credit Card but paying in Cash, switching to EFT/Ozow after a card problem, or using an In-App Wallet on a particular trip.

This does not prove how the data was generated, but the complete lack of variation is an important analytical limitation.

Business Implication

A production-level investigation of payment preferences would benefit from independent transaction information such as the preferred method, initially attempted method, failed payment attempts, successful final method and fallback method. This would allow the platform to identify payment friction and understand why riders sometimes switch channels.

Conclusion: The supplied dataset shows a 100% agreement between stated payment preference and actual payment behaviour. RQ6 therefore finds no observable difference between preferred and actual payment methods, but this perfect agreement limits the behavioural insight that can be obtained from the dataset.

7.7 RQ7 Payment Method and Trip Cancellations

Research Question: Is there a relationship between the payment method and the likelihood of a rider-initiated or driver-initiated cancellation of a trip?

The cancellation results are remarkably consistent across all five payment methods. Credit Card has the highest overall cancellation rate at 33.91%, followed by Debit Card at 33.87%, EFT/Ozow at 33.42%, In-App Wallet at 33.03% and Cash at 32.47%.

Cancellation Rates by Payment Method

Payment Method	Total Trips	Rider %	Driver %	Total Cancellation %
Credit Card	10,353	17.31%	16.60%	33.91%
Debit Card	9,863	16.83%	17.04%	33.87%
EFT/Ozow	9,704	16.71%	16.70%	33.42%
In-App Wallet	9,575	16.96%	16.07%	33.03%
Cash	10,505	16.36%	16.11%	32.47%

The difference between the highest cancellation rate, Credit Card at 33.91%, and the lowest, Cash at 32.47%, is only 1.44 percentage points. Rider- and driver-initiated cancellation rates are also closely matched within each payment method. EFT/Ozow is particularly balanced at 16.71% rider cancellations and 16.70% driver cancellations.

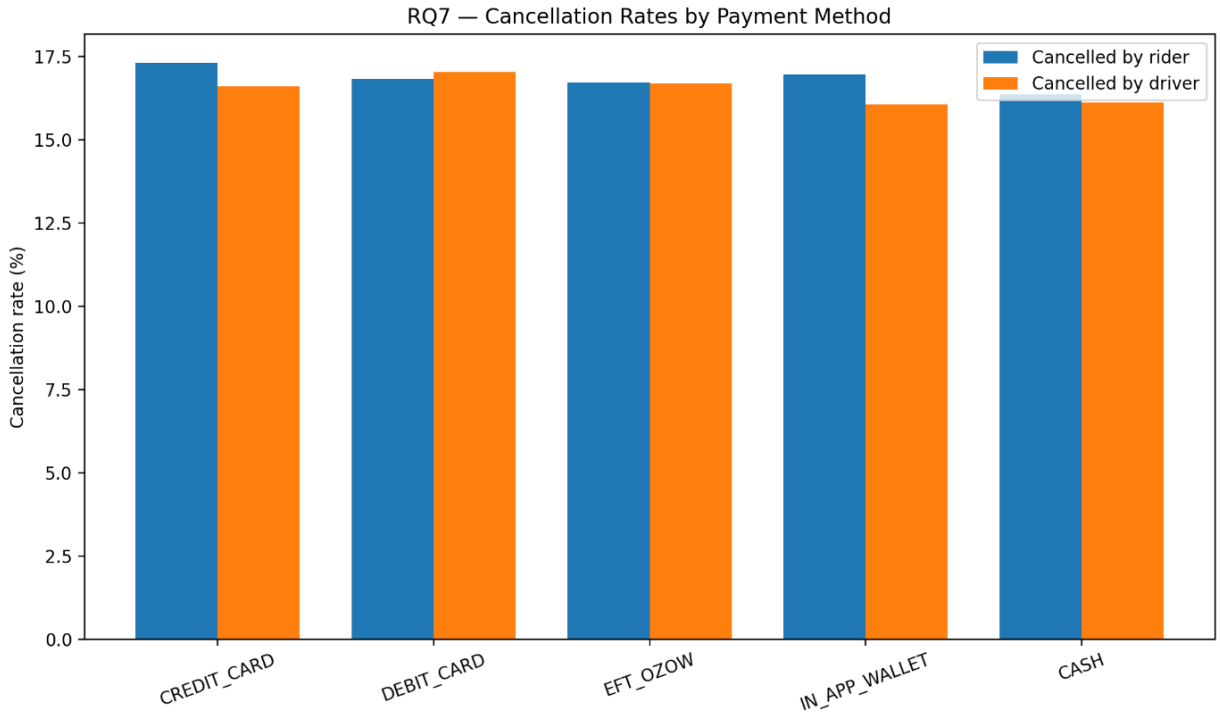


Figure 50 Cancellation Rates by Payment Method Chart

Pattern and Operational Insight

There is no strong observable association between payment method and cancellation likelihood in this dataset. If payment channel were a major driver of cancellations, substantially larger differences between payment groups would be expected.

This suggests that cancellation-reduction initiatives should focus on other trip and operational variables rather than payment channel alone. Relevant areas for further investigation include driver availability, driver distance from the rider, waiting time, traffic delays, surge-price sensitivity, rider no-shows and the recorded cancellation reason.

From an operational perspective, there is little evidence here to justify designing cancellation interventions specifically for Credit Card, Cash, EFT/Ozow or any other payment method.

Conclusion: Payment method does not appear to be a meaningful driver of cancellation behaviour. Overall cancellation rates remain between 32.47% and 33.91% across all five channels, and both rider- and driver-initiated cancellation rates remain similarly stable.

7.8 RQ8 Surge Pricing and Rider Spending

Research Question: How does surge pricing affect rider spending across different ride categories and payment methods?

RQ8 produces the clearest spending pattern of the four research questions. Across all completed trips, average fare rises substantially as the surge band increases. There were 22,393 no-surge trips, 2,530 low-surge trips, 4,977 medium-surge trips and 3,431 high-surge trips.

Average Fare by Surge Pricing Band

Surge Band	Completed Trips	Share of Completed Trips	Average Fare
No Surge	22,393	67.18%	R366.40
Low Surge	2,530	7.59%	R521.62
Medium Surge	4,977	14.93%	R641.51
High Surge	3,431	10.29%	R790.42

The transition is clear: R366.40 to R521.62 to R641.51 to R790.42. Compared with trips without surge pricing, average fare is approximately 42.4% higher under low surge, 75.1% higher under medium surge and 115.7% higher under high surge. High-surge completed trips therefore cost approximately 2.16 times as much on average as no-surge trips.

Surge Pricing Across Ride Categories

The same increasing pattern is visible in every ride category.

No-Surge vs. High-Surge Average Fare by Ride Category

Ride Category	No Surge	High Surge	Approx. Increase
Budget	R369.47	R789.43	+113.7%
Comfort	R361.36	R797.39	+120.7%
Large Capacity	R365.88	R789.86	+115.9%
Standard	R368.65	R785.06	+112.9%

The underlying SQL results show a progressive increase across the low, medium and high surge bands for all four categories rather than an increase occurring in only one category.

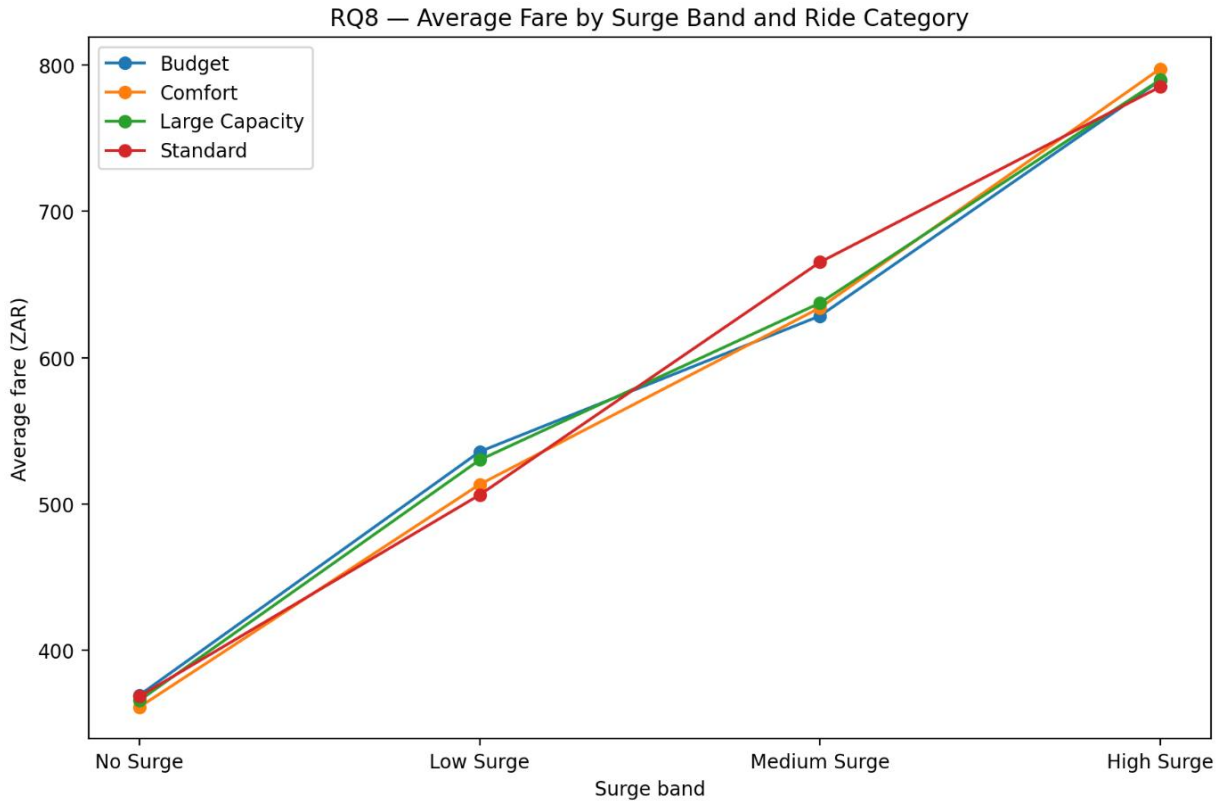


Figure 51 AVG. Fare by Surge Band and Ride Cat. Chart

An important observation is how closely the four lines follow one another. Ride category produces some variation, but the surge band has a much more visible effect on average fare than the differences between categories within the same band.

Surge Pricing Across Payment Methods

The same relationship is observed when trips are grouped according to payment method.

No-Surge vs. High-Surge Average Fare by Payment Method

Payment Method	No Surge	High Surge	Approx. Increase
Cash	R365.76	R777.92	+112.7%
Credit Card	R364.94	R782.35	+114.4%
Debit Card	R368.58	R804.64	+118.3%
EFT/Ozow	R370.80	R788.37	+112.6%
In-App Wallet	R361.99	R799.17	+120.8%

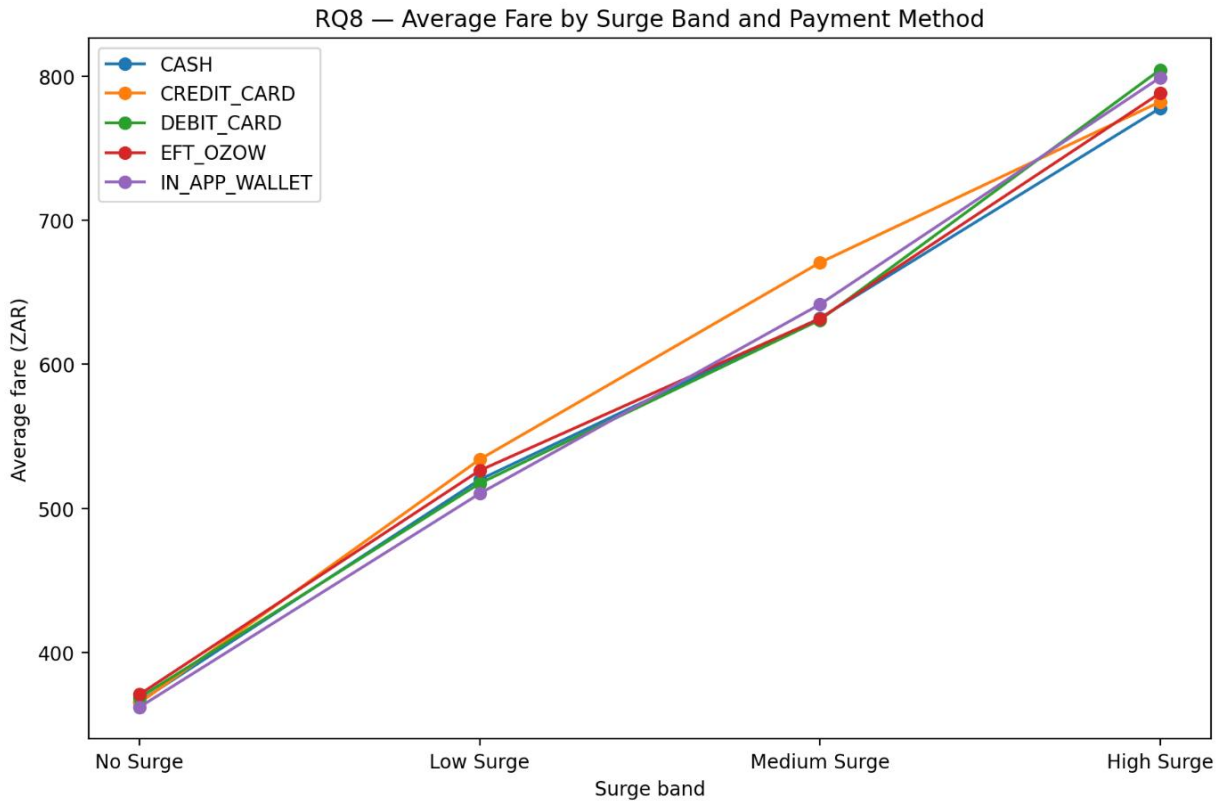


Figure 52 Avg. Fare by Surge Band and Payment Method Chart

Again, the lines follow very similar trajectories. This suggests that the surge effect is broad rather than payment-channel specific. Whether a rider pays using Cash, Credit Card, Debit Card, EFT/Ozow or In-App Wallet, higher surge levels are associated with substantially higher average fares.

Business and Consumer Insight

From a business perspective, surge pricing is an important revenue mechanism because approximately 32.82% of completed trips occur under some form of surge pricing, and those trips generate considerably higher fares per completed journey.

However, the finding must be interpreted carefully. The results show the effect of surge on completed trips. They do not reveal how many riders may have seen a high surge price and abandoned the request before a trip record was created.

The data therefore supports the conclusion that higher surge levels are associated with substantially higher rider spending when the trip is completed. It does not support the stronger conclusion that continuously increasing surge levels would always maximise overall platform revenue.

From a consumer perspective, the analysis shows that surge conditions can substantially alter trip cost regardless of ride category or payment channel.

Conclusion: RQ8 identifies a strong positive relationship between surge pricing and rider expenditure. Average fare increases from approximately R366.40 with no surge to R790.42 under high surge, and this pattern remains consistent across all four ride categories and all five payment methods.

7.9 Supporting Analysis - Satisfaction vs. Payment Method

It is clear that rider satisfaction is quite uniform across the different five payment methods analysed. On average, riders rate their drivers from 3.62 to 3.66 out of 5 depending on the payment method used.

EFT/Ozow and Debit Card payments are the two highest average ratings (3.66 and 3.65 respectively), followed by Credit Card (3.65), Cash (3.64), and In-App Wallet (3.62). The difference between the highest and lowest average rating is only 0.04, which shows the insignificance of variation in rider satisfaction in terms of payment methods used.

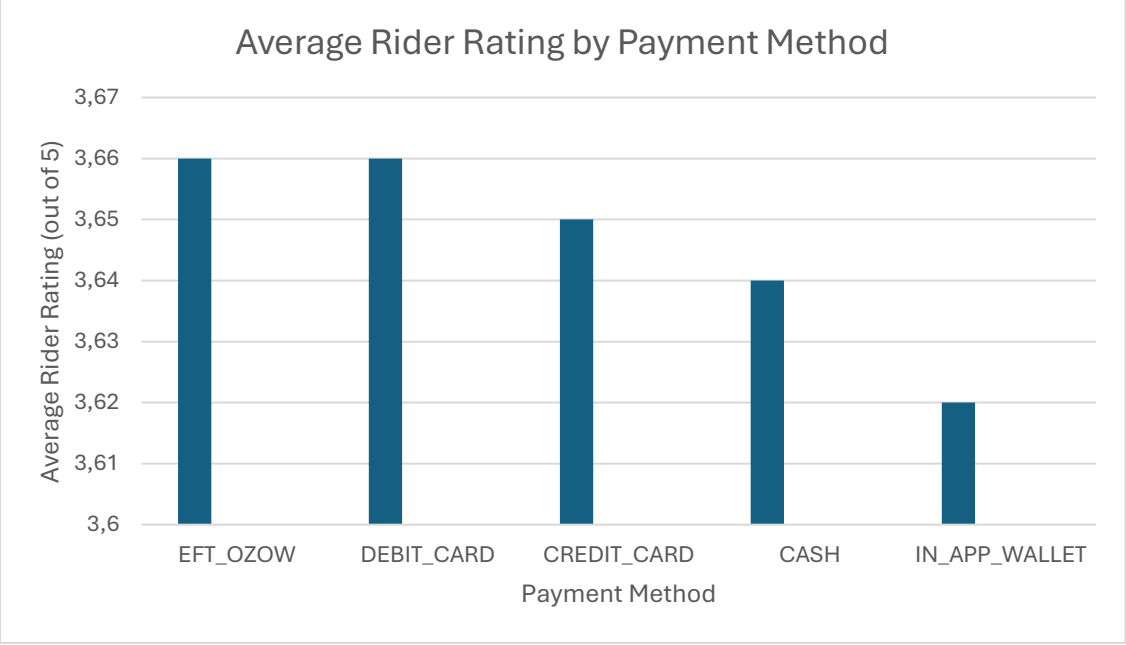


Figure 53 Avg. Rider Rating by Payment Method Chart

This figure shows the different payment methods. EFT_OZOW and DEBIT_CARD has the highest rider rating whereas IN_APP_WALLET has the lowest.

7.10 Supporting Analysis High-Value Riders

A total of 582 riders out of the 1,200 riders fall into the category of being high-value riders. High-value riders are those whose average cost per completed ride is above the average amount for all rides on the app, which is estimated at R462.91.

The rider who spends the most money is RIDER_000216 from Gauteng province and has made 27 rides with an average cost of R666.64 per ride. There are other high-value riders in Gauteng, Western Cape, Eastern Cape, KwaZulu-Natal and North West. This means that spending above the average is not confined to any single province.

From these results, one can conclude that high-value customers are present across the different geographic markets. In terms of conducting business operations, focusing on individuals who spend above the average may be a more effective approach compared to targeting the entire province. The knowledge provided by Uber or Bolt about such high-value riders may help with customer loyalty programmes.

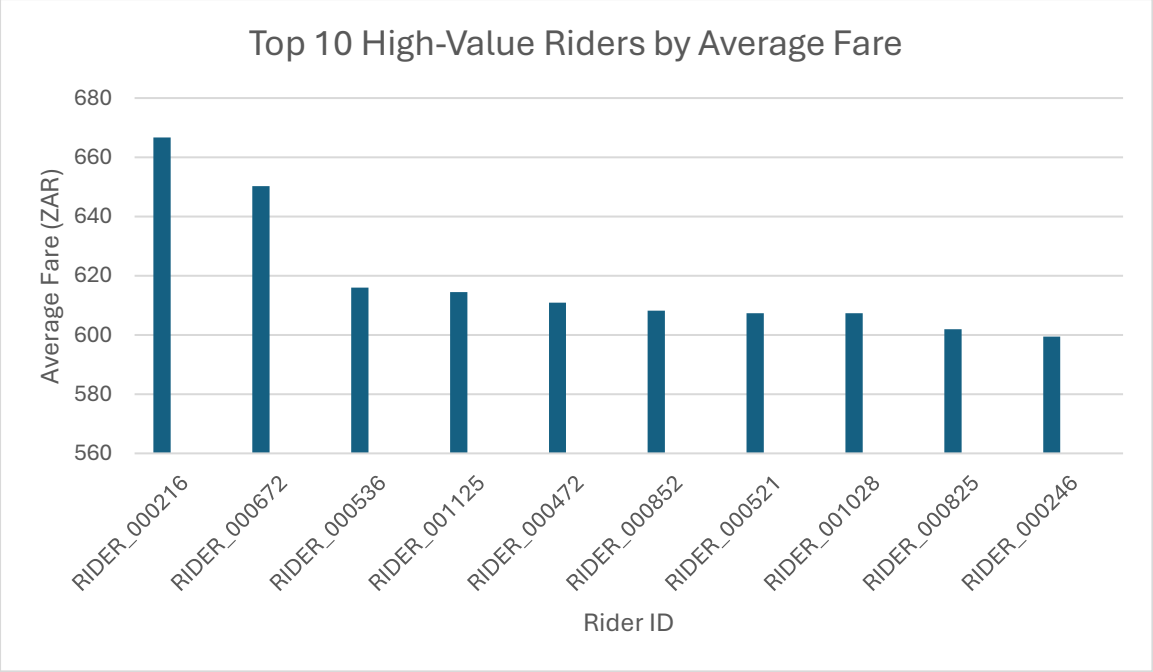


Figure 54 High-Value Riders by Avg. Fare Chart

This figure shows the top 10 high-value riders ranked by average fare per completed trip. The highest-value rider is RIDER_000216 with an average fare of R666.64, followed by RIDER_000672 at R650.21. The top 10 riders come from multiple provinces, including Gauteng,

Western Cape, Eastern Cape and KwaZulu-Natal, which suggests that high-value riders are geographically distributed rather than concentrated in one area.

7.11 Summary of Key Insights

The SQL analysis answered the eight research questions relating to rider purchasing patterns and payment channel preferences and also included two supporting analyses. Overall, the results show that rider spending in this dataset is influenced more strongly by trip-level characteristics, especially distance and surge pricing, than by province, ride category or payment method. Payment behavior is geographically varied, while cancellation and satisfaction results remain relatively stable across payment channels.

7.11.1 Summary of Findings

Analysis Area	Main Result	Business / Analytical Implication
RQ1: Provincial spending	Average fare per completed trip is very similar across provinces, from R459.19 in North West to R465.94 in KwaZulu-Natal. Gauteng has 14,850 completed trips, 44.6% of all completed trips.	Revenue differences are driven more by trip volume than by provincial fare levels.
RQ2: Trip distance	Average completed-trip distance remains in a narrow 23.8 km to 25.3 km range across province and ride-category groups.	There is little evidence in this dataset that province or ride category determines trip distance.
RQ3: Distance, duration and fare	Distance and total fare have a strong positive correlation of about r = 0.752 to 0.761. Average category fares differ by only about R5.	Distance is a much stronger spending driver than ride category in the supplied data.
RQ4: Tipping	66.4% of completed trips receive a tip. Province-level tip frequency is approximately 65.4% to 68.2%,	A broad national tipping prompt is more defensible than strongly

Analysis Area	Main Result	Business / Analytical Implication
	while the average tip on tipped trips is about R23.90.	different province-specific strategies.
RQ5: Payment adoption	All five payment methods are widely used. Cash is the largest national channel at 21.28%, but no method dominates. Provincial leaders differ.	The platform should maintain multiple payment channels and consider targeted digital payment incentives where cash use is relatively higher.
RQ6: Preferred vs actual payment	All 33,331 completed trips show the same actual payment method as the rider's stored preferred method, producing a 100% match rate.	This is an important dataset limitation and does not allow realistic payment-switching behavior to be analysed.
RQ7: Cancellations	Overall cancellation rates range only from 32.47% for Cash to 33.91% for Credit Card.	Payment method does not appear to be a meaningful cancellation driver. Operational causes should receive more attention.
RQ8: Surge pricing	Average fare rises from R366.40 with no surge to R790.42 under high surge. About 32.82% of completed trips occur under some form of surge.	Surge pricing is a major spending and revenue factor, but the dataset cannot measure riders who abandon a request before a trip record exists.
Supporting 7.9: Satisfaction	Average rider rating of drivers remains between 3.62 and 3.66 across payment methods.	There is no meaningful evidence that service satisfaction differs by payment channel.
Supporting 7.10: High-value riders	582 of 1,200 riders have an average completed trip spend above the platform average of R462.91. The highest is RIDER_000216 at R666.64 across 27 completed trips.	Customer-level loyalty targeting is more precise than treating an entire province as high-value.

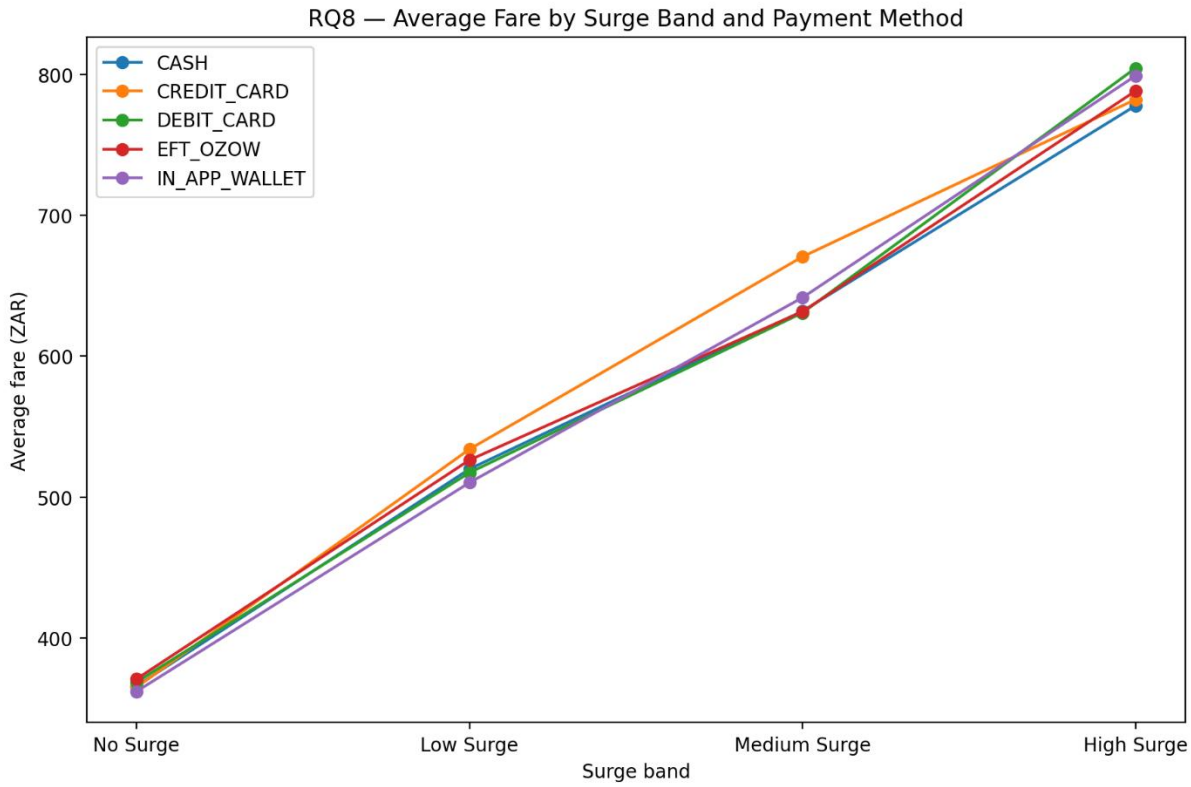


Figure 55 Avg. Fare by Surge Band and Payment Method Chart

The figure reinforces the strongest spending pattern identified in the project: surge level has a much larger effect on average fare than payment channel differences within the same surge band.

7.11.2 Overall Business Recommendations

Prioritise completed-trip volume rather than province-specific fare changes. Provincial average fares are almost identical, while trip volumes differ substantially.

Use rider-level loyalty and retention initiatives for above-average-spending customers instead of assuming that an entire province represents a high-value market.

Continue supporting all five payment channels. If the business wants to encourage digital payments, promotions can be targeted to provinces where Cash has relatively stronger adoption, particularly the Western Cape and KwaZulu-Natal.

Focus cancellation-reduction work on operational factors such as driver availability, driver distance, waiting time, traffic delays, surge sensitivity and rider no-shows rather than on payment channel.

Treat surge pricing as an important revenue lever but evaluate it together with request-level abandonment data before concluding that higher surge always improves total revenue.

Use a broad national tipping prompt or default suggestion rather than assuming that large province-specific tipping differences exist in the supplied dataset.

7.11.3 Important Data Limitations

The 100% match between preferred_payment and actual payment_method means that the dataset contains no observable payment-switching behavior. Real payment failures, fallback methods and changes between channels cannot be studied from these fields.

Trip distance and average spending are unusually uniform across provinces and ride categories. These results should be interpreted as characteristics of the supplied dataset and should not automatically be generalised to all real South African Uber or Bolt activity.

The surge analysis includes only trips that produced trip records. It cannot quantify riders who saw a surge price and abandoned the request before a trip was created.

The analysis identifies associations and patterns in the dataset. It does not by itself prove that one variable cause another.

7.11.4 Overall Conclusion

The analysis shows that rider purchasing behavior in the supplied dataset is shaped mainly by trip-level factors. Distance has a strong relationship with fare, and surge pricing produces the clearest increase in rider spending. In contrast, province and ride category show relatively small differences in average trip value. Payment channels are diverse across provinces, but payment methods are not strongly associated with cancellation rates or rider satisfaction. The results therefore support a strategy focused on increasing successful trip volume, improving operational reliability, retaining high-value riders, maintaining payment choice and applying surge pricing carefully. These conclusions should be interpreted alongside the identified data set limitations, particularly the perfect match between preferred and actual payment methods and the lack of pre-trip abandonment data.